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Optimal tax routing: network analysis of FDI diversion

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Abstract The international corporate tax system is considered a network. Just like for transportation, "shortest" paths are computed, which minimize tax payments for multinational enterprises when they repatriate profits. We include corporate income tax rates, withholding taxes on dividends, double tax treaties, and double taxation relief methods. We find that treaty shopping leads to an average potential reduction of the tax burden on repatriated dividends of about 6% points. An indicator for centrality in the tax network identifies the UK, Luxembourg, and the Netherlands as the most important conduit countries. Low-tax havens do not have a crucial role in reducing dividend repatriation taxes. By contrast, tax haven financial centres do. In the regressions we find that the centrality measures are robustly significant explanatory variables for bilateral FDI stocks. This also holds for our treaty shopping indicator.

Keywords Corporate taxation · Tax treaties · Treaty shopping · Tax havens · Shortest path

JEL Classification F23 · H25 · H26 · H87

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1 Introduction

Treaty shopping is the practice whereby multinational enterprises (MNEs), rather than investing directly in a host country, redirect the investment through a third country to take advantage of treaty provisions not found between the host and the home country of the investment (Davies 2004). These provisions are formulated in bilateral tax treaties to avoid double taxation of corporate income and to stimulate mutual foreign direct investment (FDI). These treaties are referred to as DTTs: double tax treaties. MNEs use these treaties to lower their tax burden.

Little seems to be known quantitatively about treaty shopping as no readily available measure exists for its magnitude or the subsequent reduction of the MNE's effective tax burden. In this paper we examine the extent to which treaty shopping may lead to FDI diversion and reduced repatriation tax rates on distributed dividends. This requires the construction of measures, or indicators, of treaty shopping. Circumventing the lack of direct data, we will construct these from publicly available national tax parameters. We examine whether these indicators contribute to explaining observed bilateral FDI positions.

Table 1 illustrates the worldwide pattern of FDI positions. Relatively small economies account for large shares in FDI. For example, total inward FDI stocks into the Netherlands in 2013 equalled USD 4263 billion, accounting for 15% of worldwide inward FDI stocks. In the same year the Netherlands ranked 17th in terms of GDP (between brackets) with a share of about 1% of worldwide GDP. For Luxembourg the discrepancy is even more pronounced, as it is for the British Virgin Islands, ranking 10th in outward FDI stocks.

Table 1 Top 10 of inward and outward FDI stocks in 2013. *Source*: IMF Coordinated Direct Investment Survey data, 2013 reporting countries

Country	Inward FDI		Country	Outward FDI	
	bln USD	%		bln USD	%
World	27,668	100.0	World	27,696	100.0
Netherlands (17)	4263	15.4	Netherlands (17)	5220	18.8
USA (1)	2755	10.0	USA (1)	4693	16.9
Luxembourg (73)	2466	8.9	Luxembourg (73)	2990	10.8
China (2)	2331	8.4	UK (6)	1717	6.2
UK (6)	1536	5.5	France (5)	1360	4.9
Hong Kong (41)	1120	4.3	Germany (4)	1252	4.5
Germany (4)	907	3.3	Switzerland (20)	1179	4.3
Switzerland (20)	809	2.9	Japan (3)	1118	4.0
France (5)	796	2.9	Hong Kong (41)	1098	4.0
Singapore (37)	751	2.7	British Virgin Is. (190)	1074	3.9

The totals of inward and outward stocks are not equal due to incomplete reporting and differences in registering stocks by home and host countries. Between brackets: GDP ranking (World Bank Data)

According to OECD (2013), the worldwide pattern of FDI is suggestive of international tax planning. Treaty shopping is one of the practices that MNEs use to exploit the differences in the national tax codes of different jurisdictions. Other such practices are transfer pricing, thin capitalization, and hybrid mismatches.[1]

To develop the treaty shopping indicators, we consider the international tax system as a network. We compute the optimal paths which minimize the tax burden of MNEs when they repatriate profits. The methodology is similar to determining the shortest path in a transportation network. The network consists of 108 countries, and the tax burdens are calculated from the statutory rates of corporate income taxes, withholding taxes on dividends, and double tax relief methods. Typically, bilateral DTTs lower reciprocally withholding taxes and provide for more generous relief methods (Avi-Yonah and Panayi 2011).

We use a multilateral dimension of corporate taxation for dividend flows. In this respect, we follow Barrios et al. (2012) in combining host and home country taxation, including tax treaties and double tax relief. They find that taxation of the home country, additional to that of the host country, has a significant negative impact on the location of new subsidiaries. Egger et al. (2009) construct effective tax rates between country pairs, reflecting overall host and home country taxation, and find that the bilateral effective tax rate has a negative impact on bilateral FDI stocks. However, they only construct these rates for direct routes and do not take account of treaty shopping. Their sample consists of OECD countries; for these countries the marginal and average effective tax rates (EMTR and EATR) are available. Lacking the data on effective tax rates for most of the non-OECD countries in our sample, we use statutory rates (see also Barrios et al. 2012).

We consider a profit repatriation optimization problem for a given subsidiary in a host country and the parent company in the home country. On tax-minimizing routes, which may involve FDI diversion, the tax parameters of other countries matter as well. All information is compiled and stored in a matrix describing the tax burden for incoming and outgoing dividends between each pair of the 108 countries. This matrix is entered into the Floyd–Warshall algorithm from graph theory, which efficiently performs the required minimizations. After a few minor adaptations, the algorithm computes the optimal tax routes. The difference between the tax costs on the direct route and on the optimal one yields the treaty shopping gain.

Moreover, the network approach enables us to identify countries most likely to perform the role of conduits, countries often accused of being accessories to tax avoidance by MNEs. A central position in the international tax network is a necessary condition for the role as a conduit. With a concept from graph theory, we compute measures of network centrality.

We employ these measures in the econometric analysis. We regress bilateral FDI stocks on tax variables and the indicator from the network approach on treaty shopping gain. In addition, we add the network centrality measures for the host and home country.

[1] The growing international concern over the erosion of corporate income tax bases is addressed by the base erosion and profit shifting (BEPS) initiative of the G20 and OECD (2013, 2015) and the Anti-Tax Avoidance Directive of the European Union (2016).

The FDI data are from the IMF CDIS database, and the tax parameters are obtained from different public sources. All data are for the year 2013.

Direct evidence of treaty shopping and FDI diversion is scarce for two reasons. First, the FDI data list the investment from a home country into a host country. However, this does not imply that the investment stays in the host or that the home country is the actual origin of the investment. Hence, from these data, diverted and undiverted FDI cannot be disentangled. Second, treaty shopping may be considered as treaty abuse and be illegal: the granting of (tax) treaty benefits in inappropriate circumstances (OECD 2015).[2] With the possibility of treaty shopping being illegal, direct evidence is even less likely. There are a few papers attempting to quantify treaty shopping. Weyzig (2013) makes use of micro-data of Dutch Special Purpose Entities (SPEs) from 2007. SPEs, in general, are entities with no or few employees, little or no physical presence in the host country and whose core business consists of group financing or holding activities (OECD 2013). By comparing the FDI flows via Dutch SPEs from a home to a host country to the direct FDI flows between those countries, derived from the balance of payment statistics, he estimates the share of bilateral FDI that is passing the Netherlands. This share is 6% points higher with a tax treaty route via the Netherlands compared to routes without a treaty. This is a large effect because on average 11% of worldwide FDI stocks has passed the Netherlands. The low withholding tax rates on dividends also have a significant impact on treaty shopping. Mintz and Weichenrieder (2010) construct the chains of corporate structures across various countries for German multinationals and relate these structures to the underlying fiscal motives. The level of withholding taxes is found to be important in determining which countries are used as a platform for investments.

We report three main findings. First, treaty shopping potentially leads to a significant reduction of the tax burden of MNEs. That reduction is 6% points on average. Second, the top 3 conduit countries according to our main measure of network centrality are the UK, Luxembourg, and the Netherlands. Low-tax havens, i.e. those countries with a low or zero corporate tax rate, are not relevant for dividend repatriation. Tax haven financial centres, however, are important. Third, the coefficients of the treaty shopping indicator and the centrality measures are robustly found to be significant, statistically and in terms of economic impact on bilateral FDI stocks. A 1%-point tax gain due to treaty shopping raises the bilateral FDI stock by 1.0%. An increase of one standard deviation of the centrality measure raises bilateral FDI stocks by 10–20%. These results confirm the relevance of the treaty shopping measures derived from the network approach.

We contribute to the literature with a novel and fundamentally multilateral approach that pairs a rigorous optimization framework with basic tax parameters. It generates quantitative indicators on treaty shopping in a network of 108 countries, including tax havens. With the exception of Hong (2017), we are not aware of other work using this approach and quantified at this scale.[3] Hong (2017) constructs tax-minimizing routes for a set of 70 economies, allowing a maximum of two conduit countries only.

[2] Preventing the Granting of Treaty Benefits in Inappropriate Circumstances, BEPS Action 6—2015 Final Report.

[3] An early paper of Gerard and Gillard (2004) applies a related approach to only three countries.

He focuses on the treaty-specific withholding taxes on dividends. The current paper differs by also including home country taxation. This contributes to the larger reduction in the repatriation tax burden of 6.0% points, instead of the 3.7% points of Hong (2017). Although home country taxation is not included in his network analysis, Hong (2017) does include corporate income tax rates and a dummy for foreign tax credit as determinants for bilateral FDI stocks in the regression analysis. He also introduces a variable indicating whether a direct route is optimal or not. This variable proves to be statistically significant.

An important limitation of both papers is the focus on dividends only. The network analysis takes the investment and profit repatriation decisions between a parent company and a subsidiary in another country as given. We allow for indirect financing structures by which MNEs reduce the repatriation taxation of dividends. Mintz and Weichenrieder (2010) refer to this as the treaty shopping motive for setting up conduit entities in third countries. We neglect royalty and interest payments for tax planning structures, which are instrumental in stripping corporate earnings in high-tax host countries. The emphasis in this paper is on treaty shopping for dividends, reducing withholding taxes, and in some cases the taxation in home countries. Still, the network approach offers a rich multi-country framework to investigate the dividend repatriation tax rates facing MNEs.[4]

The paper proceeds as follows. Section 2 describes the network approach to the international corporate tax system and defines the indicators. In Sect. 3 the data of the tax system are presented. Section 4 gives the effects of double tax relief methods and the DTTs on the double tax rates. The tax reduction results of treaty shopping are discussed in Sect. 5. Section 6 presents the results of the centrality indicator, identifying conduit countries. Section 7 tests these indicators with an econometric analysis. The concluding section summarizes and discusses directions for further research.

2 A network approach to international corporate taxation

This section sets out the main methodological framework. The international corporate tax system is considered as a network of countries where distance is defined as the tax burden of repatriating corporate income from one country to another country. First, it is described how these distances are composed from different international taxes, following Barrios et al. (2012). As our sample contains 108 countries, we have more than 10 thousand bilateral relations. Next, the tax burdens of these relations are discussed, involving conduit countries and treaty shopping. Third, it is shown how the tax burdens fit into efficient algorithms for computing shortest paths over the whole network.[5] Hence, the shortest paths are optimal tax routes. Fourth, a measure of network centrality is introduced to identify the most used conduit countries.

[4] We do use the withholding tax rates on royalties as explanatory variables in the regressions. Due to the strong correlation between the withholding tax rates for royalties and interest, we only include the former.

[5] For further discussion of the network algorithm and its tax application, see Appendix C1.

2.1 Tax burden, treaty shopping, and a network algorithm

Consider a multinational with a subsidiary in a host country S and a parent company in home country P. Both countries may tax the income of the subsidiary. First, the host country levies a corporate income tax (CIT), at a rate t_S. Next, the host country may also levy a general non-resident withholding tax on the dividends of the subsidiary. The tax is denoted with w_S. However, the host and home country may have signed a tax treaty and a preferential rate $w_{SP} \leq w_S$ may apply. The home country may tax the foreign-source income at its CIT rate, t_P.

The tax code of the home country may contain provisions to avoid double taxation. For instance, it may have a dividend participation exemption: under certain conditions all, or part, of the foreign-source dividend income is exempted from the corporate income tax. These conditions typically require a minimum share in the subsidiary and a minimum number of years that the shares are held. Some countries do not apply tax relief to dividend from low-tax countries or tax havens (CFC rules). Apart from *exemption* two other methods of double tax relief are taken into account: *deduction* and *credits*.[6] Deduction is the most modest relief method where no taxes need to be paid over the taxes already paid. The latter are deducted from the tax base in the home country. With the credit system the tax base is the income of the subsidiary, but the corporate income tax is credited with taxes paid in the host country.[7] Excess credit is not restituted. The credit method is less generous than exemption, but more so than deduction.

Let rm_P be the general double tax *r*elief *m*ethod applied by home country P. The tax treaties the country has signed may contain agreements to provide more generous double tax relief to treaty partner S. Thus, also the relief methods have a double country dimension: rm_{SP}. The resulting combined tax rates after applying the relief method are fully in line with Barrios et al. (2012).

$$t^e_{SP}(\text{deduction}) = 1 - (1 - t_S)(1 - w_{SP})(1 - t_P)$$

$$t^e_{SP}(\text{credit}) = \max\{1 - (1 - t_S)(1 - w_{SP}),\ t_P\}$$

$$t^e_{SP}(\text{exemption}) = 1 - (1 - t_S)(1 - w_{SP})$$

Next, we define the direct tax distance or burden d_{SP} between host S and home country P based on the relevant double taxes; the withholding tax rate and the CIT of the parent. The CIT of the host country is excluded, because this tax is always paid, irrespective of the relief method and therefore irrelevant for our treaty shopping analysis. Depending on the tax relief method, the tax distances are defined as follows.

$$d_{SP}(\text{deduction}) = 1 - (1 - w_{SP})(1 - t_P)$$

[6] Thus no-relief-at-all, which does occur sparingly, is ignored.

[7] With an indirect tax credit both the host corporate income tax and the withholding tax are credited. With a direct tax credit only the withholding tax can be credited. We ignore the latter.

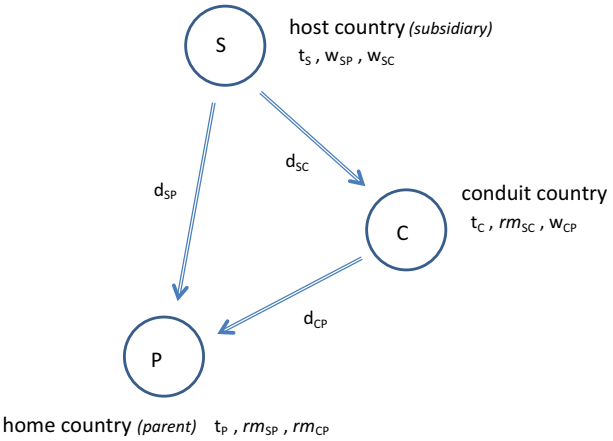


Fig. 1 Treaty shopping—one conduit country

$$d_{SP}(\text{credit}) = \max\{w_{SP},\ (t_P - t_S)/(1 - t_S)\}$$

$$d_{SP}(\text{exemption}) = w_{SP}$$

The tax distances represent the double taxation of profits of the subsidiary in the host country. Full double taxation for a given pair would be $d_{SP}^{\max} = 1-(1-w_S)(1-t_P)$, with deduction as the standard relief method and application of the general withholding tax rate. Double tax relief then is the difference between the full and the actual tax distance, $d_{SP}^{\max} - d_{SP}$.

We refer to the tax distances d_{SP} as the bilateral direct dividend repatriation tax rates.[8]

Now consider the possibility of indirect repatriation of dividends, i.e. through a third, or conduit, country C; see Fig. 1. It is rational for the MNE to choose the indirect route over the direct route, *ceteris paribus*, when the tax burden is lower, i.e. $1-(1-d_{SC})(1-d_{CP}) < d_{SP}$. As the CIT of host S is to be paid in both cases, it does not matter for the (absolute) comparison. The conduit country functions both as an intermediate host and as an intermediate home country.

The CIT of an intermediate country is relevant when the next intermediate, or final, home country at a tax route applies the credit method. Then it may not be clear which taxes can be credited; all the taxes of the preceding part of the tax route, or just the taxes paid in the previous country? In these conduit situations we take the rate of the world average corporate income tax to be credited.[9] This weighted average excludes the CIT rate of the country involved. The withholding taxes of the previous country are always taken into account and are credited where required.

[8] Altshuler and Grubert (2002) also use the term 'repatriation tax'. They analyse the repatriation decision, which we take as given.

[9] Alternative approaches are discussed in Appendix C1.

As an example of an indirect route, assume that home country P applies the deduction method and conduit country C exempts foreign-source dividend income. The treaty shopping condition translates to $w_{SC} + w_{CP} - w_{SC}w_{CP} < w_{SP}$; the combined withholding taxes on the indirect route must be less than on the direct route. This is only possible when the withholding tax rate to the conduit country is lower than the one to the parent, $w_{SC} < w_{SP}$.[10]

The tax burden of an indirect tax route with a single conduit country is the usual combined effective rate of two tax rates. But a shortest path may go beyond the triangles above. Thus, in general, for any tax route, with an initial host $k = 1$ and final destination $k = n$, the total tax distance equals $1 - \prod_{k=2}^{n}(1 - d_{k-1,k})$. Clearly, the order of the bilateral tax distances in the computation does not matter. This characteristic allows the use of standard and efficient algorithms to determine the tax burden of the optimal route between all pairs of countries in the network; or rather the minimum tax costs of repatriating dividends for all country pairs. We use the elegant Floyd–Warshall algorithm.[11] It stepwise builds up the matrix of shortest tax distances by consecutively adding and evaluating a new country, in arbitrary order, as an intermediate or conduit country. The efficiency of the algorithm is important as the number of possible routes over a network is huge.[12] The algorithm generates the matrix of optimal tax rates in repatriating dividends from all host countries to all home countries, s_{ij}. A first indicator from the network approach is the treaty shopping gain, defined at the bilateral level as the difference between the direct and the optimal tax costs, $d_{SP} - s_{SP}$.

2.2 A centrality measure for conduit countries

The lowest tax burden for a particular country pair may be incurred on the direct route or on an indirect one. In the latter case the route contains intermediate countries. We identify these conduit countries using a centrality measure from network theory: *betweenness*. Given all optimal paths between all country pairs in the network, betweenness centrality is defined as follows: for a given country, count the number of times it is on an optimal path from S to P. Divide this number by the total number of optimal paths between S and P. Then sum these fractions over all pairs S and P, excluding those pairs where the given country is the initial host or home.

We have performed the summation over country pairs for the betweenness centrality measure with double GDP weights. These weights are taken as a proxy for the relative size of undiverted dividend flows, which ideally would have been used. However, these are unknown. Although the OECD has some sparse data on reported bilateral dividend payments, it must be realized that these data are burdened with the central

[10] Given that the undiverted investment also would have taken place without the treaty shopping, this implies that the host country loses tax revenue. This is usually the case, and the OECD concludes that treaty shopping is a harmful tax practice (OECD 1998). IMF (2014) warns that treaty shopping is particularly harmful for developing countries.

[11] See, for instance, Minieka (1978).

[12] For a simple network that has ten countries and is complete, meaning that all direct pairwise connections exist, there are almost 10 million simple routes.

Table 2 Summary of tax variables

Symbol/definition	Variable name	Text
t_i	CIT_i	Corporate income tax rate
w_i	WHT_i^{div}	Default (general) withholding tax rate on dividends
	WHT_i^{roy}	Default (general) withholding tax rate on royalties
w_{ij}	$BWHT_{ij}$	Bilateral withholding tax rate (on dividends)
d_{ij}	$DRCT_{ij}$	Bilateral direct dividend repatriation tax rate (i.e. on direct routes)
$d_{ij} - w_{ij}$	$RCIT_{ij}$	Remaining cit rate in home country (on direct routes)
s_{ij}	$OPTM_{ij}$	Bilateral optimal dividend repatriation tax rate (i.e. on optimal routes)
$d_{ij} - s_{ij}$	$GAIN_{ij}$	Treaty shopping gain/treaty shopping indicator
	$CENT_i$	Centrality measure

issue of this article, the diversion for fiscal reasons. The same applies to FDI data.[13] Not weighting the country pairs in the measure implicitly assumes equal bilateral flows, which would be highly unrealistic given the different countries in the network. We will show the results with GDP-based weights first, but present unweighted centrality as an alternative measure. Whatever weights are used, the betweenness centrality of a country can be interpreted as the share of total worldwide undiverted flows that pass through it. This is shown in Appendix C2, as is the construction of the weights.

All results on treaty shopping gain and the tax burden are derived at the level of the country pair. To present aggregate results the double GDP weights are used. We have country level results for the host (outbound dividends) and home countries (inbound dividends). Table 2 presents a summary of tax variables, also stating the names used in the regression analysis.

3 Data: national tax parameters

The selection of 108 jurisdictions contains all high- and upper-middle-income economies[14] for which sufficient tax data are available. This is augmented with lower-middle-income countries with relatively large economies, such as India, Indonesia, and the Philippines. The selection covers almost 95% of worldwide GDP in 2013. The full list is found in Appendix A1. The selection also includes many jurisdictions classified as a tax haven, because the latter are usually small and affluent; see, for instance, Dharmapala and Hines (2009). The importance of including tax havens is evident: they are likely conduit countries if only for their characteristic of low or zero taxes (OECD 1998). Avoiding precise definitions we refer to the list of Gravelle (2013) as

[13] In an alternative exercise, in collaboration with the OECD, bilateral weights were constructed based on sparsely available but reported bilateral dividend flows and FDI stocks; see Van 't Riet et al. (2015).

[14] World Bank Atlas method, based on 2013 GNI per capita data.

a benchmark for tax havens. In the end, we classify 26 countries in our list of 108 countries as tax havens. As a subgroup we identify 10 low-tax havens with a CIT rate of 12.5% or less, which includes Bermuda, the Cayman Islands, and the British Virgin Islands. Another subgroup of five tax havens, for which we will find different results, are financial centres, Ireland, Jordan, Luxembourg, Singapore, and Switzerland (see Appendix A2).

The tax data are mainly obtained from the Worldwide Corporate Tax Guide from EY (2013). For each country, we have data on the corporate income tax rate, the general rate of the withholding tax on dividends, the general double tax relief method, possibly the more lenient tax relief method for treaty partners, and the treaty dividend withholding tax rates. For the dividend tax rates, we choose normally the lowest rate which is often conditional on a substantial participation in the daughter company.[15] Quite often this is 10–25% of the stocks, but sometimes the lowest tax rate applies only if the parent owns the majority of the stocks. Although the data have been cross-checked with other public sources,[16] still some errors and omissions are expected to remain. In addition, choices and interpretations are unavoidable as tax codes contain different rates and provisions that apply under different conditions, which may involve legal structures, the level of corporate income, the industry, ownership shares, etc. Our choices and best knowledge are found in Appendix A1 (except for the treaty withholding tax rates).

Statutory rates of corporate income taxation have been used, where applicable including local taxes.[17] We ignore the possibilities to reduce the tax base in the host countries and the effective tax rates for three reasons. First, as we are mainly interested in the routing decision of repatriating income given the ultimate host and home country, the deduction possibilities of these two countries may apply whatever the route chosen; statutory or effective rates then do not affect comparison. Second, reduced taxation in the host increases the profit base to be repatriated and hence the scope for further tax reduction. This may be relevant when deduction or the credit method is involved. However, when intermediate or final home countries apply dividend exemption as many countries do, neither effective (as in EATR) nor statutory rates are relevant. The tax minimization will make sure that the chosen routes include as much as possible countries with the dividend participation exemption. A third and practical reason for not using effective tax rates is that these are simply not available for most countries in our selection.

The CIT rates are listed in the first column of Table 3 for a selected number of countries and in Appendix A1 for all countries. When foreign-source income is exempted from corporate taxation (xmp), the tax rate is irrelevant, for the other double tax relief methods it is not. The general tax relief method is indicated in the second column. Countries may provide more generous relief for foreign-source dividends coming from tax treaty partners, such as Canada; see column (3).

The CFC column (controlled foreign corporation) indicates when a country applies anti-abuse provisions to counter tax deferral and avoidance through artificial foreign

[15] We have ignored lowest tax rates which only apply to non-profit organizations, such as pension funds and government institutions.

[16] For instance, Deloitte (2013) and Loyens and Loeff (2013).

[17] OECD Tax Database and KPMG Tax Tools and Resources.

Table 3 Tax data 2013—selected countries

Country (unit)	CIT rate (%) (1)	Relief method (2)	Treaty relief method (3)	CFC rule (dummy) (4)	WHT^{div} (%) (5)	DTT (no.) (6)	Tax haven (dummy) (7)	GDP weight (%) (8)
Bermuda	0.0	xmp		0	0	0	1	0.01
Brazil	34.0	crd		1	15	35	0	2.97
Canada	26.3	crd	xmp	0	25	75	0	1.88
China	25.0	crd		1	10	64	0	15.66
France	34.3	xmp*		1	30	80	0	2.84
Germany	30.2	xmp*		1	25	71	0	4.04
Hong Kong	16.5	xmp		0	0	14	1	0.47
Japan	37.0	xmp*		0	20	48	0	5.84
Luxembourg	29.2	xmp		0	15	56	1#	0.05
Malta	35.0	xmp		0	0	40	1	0.01
Netherlands	25.0	xmp		0	15	72	0	0.89
Switzerland	21.1	xmp		0	35	71	1#	0.46
United Arab Emirates	0.0	xmp		0	0	21	0	0.34
UK	23.0	xmp		1	0	51	0	2.95
USA	39.1	crd		1	30	54	0	19.79

crd credit system, *xmp* exemption, *ded* deduction system, *xmp** exemption up to 95%
#Tax haven financial centre

entities. For these countries we interpret tax relief for dividends from tax havens as deduction. This is the only instance where tax havens have a specific role in the network analysis. They are listed in column (7). The five tax haven financial centres are excluded from CFC treatment.

Tax havens are often low-tax countries, as is the case for Bermuda. Malta, listed as a tax haven, has a high nominal corporate income tax of 35%. For holding companies, however, this is irrelevant as Malta applies a dividend participation exemption. Tax havens tend to have a zero withholding tax on dividends. The general rates of this tax are found in the fifth column of Table 3 and Appendix A1. The sixth column indicates the number of bilateral tax treaties a country has with partners within the selection of 108 jurisdictions.[18]

The bilateral withholding tax rates from the DTTs imply a matrix structure for the tax data; see Table 4. An important multilateral tax treaty is the Parent–Subsidiary directive of the EU.[19] This stipulates intra-EU withholding tax rates of zero and dividend participation exemption.

[18] The number refers to treaties where at least one of the bilateral withholding tax rates has been reduced.

[19] European Union (1990). Next to the 27 EU member countries in 2013, also Iceland, Norway, and Switzerland are included.

Table 4 Bilateral dividend withholding tax matrix 2013—selected countries

From \ To	Bermuda	China	Germany	Malta	NLD	Nigeria	USA	General
Bermuda	–	0	0	0	0	0	0	0
China	10	–	10	5	10	7.5	10	10
Germany	25	10	–	0	0	25	5	25
Malta	0	0	0	–	0	0	0	0
Netherlands	15	10	0	0	–	12.5	0	15
Nigeria	10	7.5	10	10	7.5	–	10	10
USA	30	10	0	5	0	30	–	30

4 Double taxation and double tax relief

The bilateral direct dividend repatriation tax rates combine the withholding tax on dividends and taxation in the home country, as explained in Sect. 2. This section presents descriptive statistics on double tax relief.

A multinational could face double taxation when repatriating profits of a foreign subsidiary. Taking the CIT of the host country as given, there remains in theory about 41% additional, i.e. double, taxation.[20] In practice double tax relief methods and tax treaties reduce or even eliminate double taxation; the combined effect is that there remains a world average double taxation of 12%. We conclude that double tax relief methods and tax treaties do what they are supposed to do: they reduce double taxation substantially, by almost 30% points.

Double taxation varies by country. We average the bilateral direct dividend repatriation tax rates for the countries as home or as host of the investments. Low average tax rates for inbound dividends make a country attractive for corporate residence. The Netherlands and Finland head the ranking of countries in this respect, with average rates of 3.4 and 3.7%; see Table 5. By contrast, the average rate for the USA is 16.7%. This implies that when US corporations would switch their legal residence to one of these European countries they could reduce the tax burden on dividends from all over the world by more than 10% points. Ireland, often a candidate country for such a tax inversion, has an average repatriation tax rate of 5.6% and ranks 11th.[21]

Exemption contributes to a low average inbound rate, certainly compared to countries with only credits for taxes already paid. The number of bilateral tax treaties also matters for the inbound rates as they stipulate reduced withholding taxes for the host country of the repatriated profits. The low-tax havens in general have no or few double tax treaties and therefore do not rank high. This is markedly different for the tax haven financial centres.

[20] World average taxation equals $1-(1-0.17)(1-0.29)\cong 0.41$.

[21] An example of a tax inversion was the planned 160 billion USD merger of the pharmaceutical companies Pfizer and Allergan, announced in November 2015. Pfizer is American but the legal residence of Allergan is in Ireland, which is also where the new company was supposed to reside. Measures in April 2016 taken by the US Treasury made the tax benefits of the merger uncertain and the merger plan was abandoned. US tax reform plans in 2017 would remove the tax incentive for inversion.

Table 5 Average direct repatriation tax rates for inward dividends—selected countries

Country	Relief method	CIT	DTT (no.)	Avg. rate	Rank
Netherlands	Exemption	25.0	74	3.39	1
Finland	Credit[#]	24.5	59	3.65	2
UK	Exemption	23.0	51	3.75	3
Luxembourg	Exemption	29.2	57	4.03	5
Switzerland	Exemption	21.0	71	4.93	9
Ireland	Credit	12.5	53	5.58	11
Canada	Credit[#]	26.3	75	6.40	18
USA	Credit	39.1	54	16.67	64
Bahamas	Exemption	0.0	0	17.39	75
World average		29.2		11.88	

[#]Credit system but exemption for treaty partners

As an illustration we present two examples of dividend repatriation tax rates. First consider repatriating dividends from a subsidiary in the USA to a mother company in China. The general non-resident withholding tax of the USA on dividends is 30%, and the CIT of China is 25%. The double taxation would be 47.5%.[22] But China applies the credit method as double tax relief: the full credit leaves a double taxation of 30%. The tax treaty of the USA and China stipulates a withholding tax of 10%; see Table 3. So, the repatriation tax rate is 10% (*DRCT*). Next, consider repatriation of dividends from China to the USA. The withholding tax in China on dividends to the USA is 10%; see Table 3. The USA also has a credit system. The CIT rate of 25% and the withholding tax rate in China do not exceed the CIT of the USA. The repatriation tax rate therefore is equal to the difference between the CIT rates, expressed on the tax base after the CIT in China, amounting to 18.8%.[23] Below we will discover that this rate cannot be lowered by treaty shopping.

5 The gains of treaty shopping

So far only direct routes between the host and home countries were considered, but firms also use indirect routes for investment and dividend repatriation. By establishing conduit entities in third countries, multinationals can lower their tax burden. This section computes the gains of treaty shopping. We find that for 67% of all country pairs this gain exists.

Treaty shopping involves deliberate diversion of investment, which will not always take place. The use of tax treaties could be bounded by the limitation of benefits articles in the treaties, although it is not very clear whether these limitations are very effective. Therefore, the gains are potential reductions. We find that the overall gain by treaty

[22] $0.475 = 1 - (1 - 0.30) * (1 - 0.25)$.

[23] $0.188 = (0.391 - 0.25)/(1 - 0.25)$; see Sect. 3 for the formula of this tax rate.

Table 6 World average dividend repatriation tax rates (%) and the distribution of the double tax revenue

	Direct	Optimal
WTH div	7.7	2.1
CIT home	4.5	3.8
Double	11.9	5.9
Host	7.7	2.1
Conduit		0.3
Home	4.2	3.4

shopping is 6% points. This lowers the worldwide average double taxation from 12 to 6%. This finding suggests that treaty shopping is a relevant mechanism for lowering the dividend repatriation tax.

Treaty shopping lowers the repatriation tax rates for two reasons. The first is that firms benefit from routes with lower withholding taxes, explaining the major part of the tax reduction. The second reason is that firms choose routes such that they benefit from more generous double tax relief methods in the home country. The latter mechanism is reflected in the fall of the average rate for the CIT of the home country; see the third line of Table 6. Table 6 also presents the split of the double tax revenue over the host, conduit and home countries. Only 0.3% of worldwide repatriated dividend flows is cashed by the tax authorities of conduit countries when the multinationals use optimal routes. Van 't Riet and Lejour (2014) show that for individual countries the consequences of indirect routing on tax revenues diverge widely.

As we did for the direct tax routes, country average repatriation tax rates are computed, for both inbound and outbound dividend flows. These results are given in Appendices B1 and B2. Starting with the tax rates for incoming dividends, Fig. 2 demonstrates the impact of treaty shopping on the average double tax rates of the home countries. The top, dashed, line shows the repatriation tax rates for direct routes, discussed in Sect. 4. The solid line below shows the optimal tax rates including treaty shopping. For both lines the countries have been ordered, from left to right, in ascending rates. As these two rankings are different, countries have different positions at both lines, as is indicated for the USA.

Treaty shopping lowers the floor in the average dividend repatriation tax rates. There is a distinct group of 82 jurisdictions with an average optimal tax rate of 1.8%, or even lower, on inbound dividends. The EU members are all in this group as they can transfer dividends within the EU without any tax cost because of the Parent-Subsidiary directive. The USA, for instance, is not part of this group. It applies the credit method, instead of exemption, and levies a high CIT rate. There are no detours to avoid this, given that repatriation of the foreign-source income is assumed, as we do not consider deferral of taxation.

A similar pattern can be seen for host countries and the average repatriation tax rates on outbound dividends. Here the MNEs face a tax rate of 5.5% or lower when repatriating profits from another group of 82 countries (68 of them identical to those 82 for inbound flows, explained below). Again treaty shopping seems to equalize

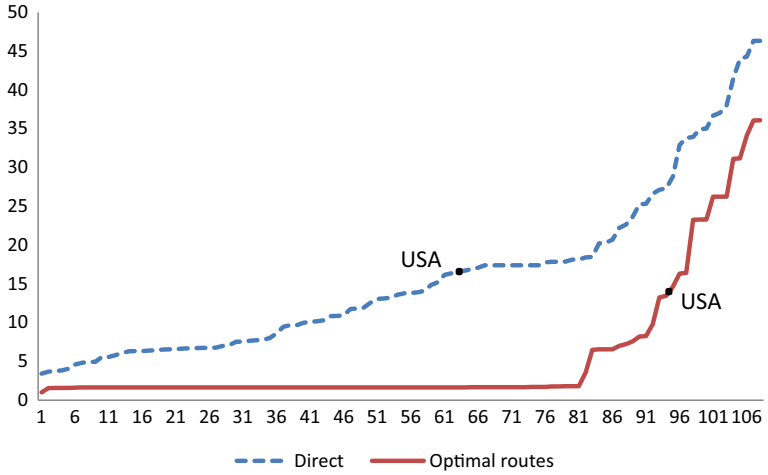


Fig. 2 Rankings of average inbound dividend repatriation tax rates

the dividend repatriation tax rates for a large group of countries including again the EU members. Tax havens are also within this cluster as they often levy no withholding taxes. Countries like Canada, China, Japan, and the Russian Federation have higher dividend repatriation tax rates on outbound dividends, because they always levy a withholding tax rate of at least 5%, even to their most favoured treaty partners.

The country averages with a lower floor in the repatriation tax rates are best understood considering bilateral rates. At the bilateral level this floor is rock bottom: zero repatriation tax rates. Even without treaty shopping non-taxed repatriation of profits exists. The initial tax distance matrix contains 2240 cells with value zero.[24] Treaty shopping increases the number of zero cells to 6212, which is 54% of all country pairs. These zero tax rates imply that within the international tax network there is a strongly connected component of 68 countries. This group is characterized by tax free dividend flows between all these countries.

A treaty shopping gain is not possible on direct routes that are already tax free; this holds for 19.4% of all country pairs; see Table 7. For another 15.5% no gain can be realized either. Complete elimination of the tax burden is possible for 34.4% of the pairs, and for 32.7% there is a partial treaty shopping gain. The latter two categories make up the two-thirds of all pairs for which an indirect repatriation tax route is optimal.

The average treaty shopping gain for individual countries is given in Appendices B1 and B2 for inbound and outbound dividends, respectively. The tax reduction by treaty shopping for MNEs residing in the Netherlands is small because the average repatriation rate on direct routes is already low. The opposite holds for the gains on outbound dividends from low-tax havens. As the average direct repatriation tax rates are high for these jurisdictions, treaty shopping allows for a reduction of more than 20%.

[24] The EU's Parent–Subsidiary directive alone is responsible for $(27+3) * (27+3-1) = 870$ zero cells.

Table 7 Distribution over country pairs of treaty shopping gain

	No. pairs	%	
No gain possible	2240	19.4	Direct routes tax free
No gain	1564	15.5	Optimal tax is direct tax, both positive
Partial gain	3780	32.7	Optimal tax lower than direct tax, both positive
Full gain	3972	34.4	Optimal route tax free, but direct route not

Table 8 Top 10 conduit countries—betweenness centrality

	Country	WHT^{div} (%)	DTT (no.)	BTWNS (%)
1	UK	0	51	13.4
2	Luxembourg	15	57	8.4
3	Netherlands	15	74	7.7
4	Estonia	0	36	6.7
5	Hungary	0	47	6.2
6	Singapore	0	40	6.1
7	Ireland	20	53	5.6
8	Slovak Republic	0	42	5.3
9	Finland	24.5	59	4.7
10	Cyprus	0	35	4.5

For dividends repatriated from China to the USA, no treaty shopping gain is possible; the 18.8% repatriation tax rate on the direct route remains. For dividends in the opposite direction, the result is diametrical; the treaty shopping gain completely eliminates the repatriation tax rate. There are 12 countries which can receive dividends from the USA without additional taxation. Next there is a group of 22 countries from which dividends to China are not taxed. The intersection of these groups consists of two countries. Hence, there are two optimal routes between the USA and China with exactly one conduit country with zero dividend repatriation taxes. These countries are the UK and Luxembourg which frequently are used as conduit countries, as is shown below.

6 Tax network centrality results: identifying conduit countries

In this section we identify the countries most often used for channelling FDI, and being on optimal dividend repatriation routes. We first discuss the results of our main centrality measure. Next we present some alternative centrality measures and establish the robustness of the results. Table 8 gives the top 10 country ranking of betweenness centrality. The full ranking can be found in Appendix B3.

The value of the betweenness measure (BTWNS) is a weighted fraction: the UK is on 13.4% of the optimal tax routes of the world average country pair. The UK

ranks high because it is an EU member and it levies no non-resident withholding tax on dividends (WHT^{div}). These two characteristics it has in common with Cyprus, Estonia, Hungary, and the Slovak Republic, all in the top 10. But the UK has signed more bilateral tax treaties, and multinationals face on average a lower withholding tax on incoming dividends. The impact of a zero withholding tax rate on this ranking is evident. Luxembourg, the Netherlands, Ireland and Finland do levy a withholding tax on dividends, but have agreed to lower or to eliminate the rate in many DTTs.[25] Singapore is the only non-EU country in the top 10.

Most of the island low-tax havens also do not levy a non-resident withholding tax on dividends. Nevertheless, with the exception of Cyprus, they do not rank high on the centrality measure. This is caused by the fact that they have no or only a few DTTs. Low-tax havens do not significantly contribute to the conduit function for repatriating dividends. This is because the other countries apply relatively high withholding taxes on profit flows towards them. Tax haven financial centres rank higher; Luxembourg, Singapore, and Ireland are in the top 10.

Although conduit countries are important for diverting FDI and optimizing the taxes on dividend flows, their impact on the average repatriation tax rate is very modest. If we exclude the top 10 conduit countries from Table 7 in the network analysis the world average dividend repatriation tax rate is only raised from 5.9 to 6.0%. The elimination of a group of countries as conduits has little impact as long as there remains a large group of well-connected countries. This seems to imply that large-scale international cooperation would be required to combat treaty shopping.

6.1 Multiplicity of shortest paths and alternative centrality measures

Centrality measures are calculated using all optimal paths. The total number of optimal paths can be much larger than the number of country pairs. In fact, multiplicity of optimal paths in the international corporate tax network is abundant. A potential source of huge multiplicity is direct routes with a zero tax rate; no double taxation at all on dividend repatriation which is the case for 2240 country pairs. A consequence of these zeros is that there are costless detours possible which are also optimal routes. However, in practice firms face costs setting up a holding, even if it is a shell company. The zero-cost detours have been countered by introducing a small penalty, initially epsilon, for each additional conduit country on a route. This reduces the average multiplicity, but it is still about 5.5 paths per country pair.

Another reason for multiplicity is that multinationals may prefer routes with slightly higher costs than the strictly optimal routes because of non-tax characteristics of the conduit countries. These may include the quality of the financial

[25] Luxembourg and the Netherlands are attractive for other tax reasons too. Liquidation of a company in Luxembourg is treated as a capital transaction and is not subject to a dividend withholding tax. The Netherlands has a general rate of zero on royalties and interest. Such characteristics have not been taken into account in the analysis. To avoid arbitrariness we stick to the bare tax parameters.

Table 9 Distribution of lengths of shortest paths

		Total paths	Multiplicity	Length = no. of conduit countries				
				0	1	2	3	4
Strict	Number	63,424	5.5	3804	50,664	8683	273	0
	%			6.0	79.9	13.7	0.4	0
Range	Number	1,050,030	90.9	3825	50,758	729,764	256,860	8823
	%			0.4	4.8	69.5	24.5	0.8

sector and government institutions; see Dyreng et al. (2015). We have allowed for a half percentage point on top of the optimal tax rate of the strictly optimal paths. This, however, reintroduces the possibility of zero-cost detours in the optimal routes. Hence, we raise the penalty to slightly below the range. This penalty, modelled as an additional tax, could represent the cost of setting up a conduit entity in a new country. Although the half percentage point is to some extent arbitrary, experimenting with combinations of the range and penalty shows a stable ranking of conduit countries.[26] Thus, paths within this range are considered as optimal paths and are included in computing the centrality measures. The average multiplicity per pair increases to 91 different relevant paths. The distribution of lengths of strictest optimal paths and those within a range of a half percentage point is given in Table 9. Length is denoted in number of conduit countries.

For 3804 country pairs, about 33% of all pairs, the direct route is among the relevant paths. These are 6% of all strictly optimal paths. In about 80% of the optimal paths there is exactly one conduit country. With a range of optimal paths, this is different. The extra paths on top of the strictly optimal paths are those with two or three countries. The maximum number of conduit countries encountered is 4. For most country pairs, however, the shortest paths contain at most two conduits. The two corresponding centrality rankings, i.e. for strictly optimal and those within range, are found not to be very different.

We find two strictest optimal paths for the USA to China, each with exactly one conduit, the UK and Luxembourg. With the range, however, there are 55 relevant paths added, all with two conduits, among them again the UK and Luxembourg, but also the Netherlands, Ireland and Finland.

Up to now we have defined two centrality measures: the one with the range of a half percentage point and the one without, respectively, *Betweenness* (BTWNS) and *Strict* (STRCT). A next centrality measure counts whether a jurisdiction is at all on an optimal path of a pair, i.e. *Occurrence* (OCCUR). This is a binary indicator instead of a fraction. By country the indicators are again summed over the pairs. When there would be a unique optimal path for each pair the two measures

[26] We take the ranking with range = 0.005 and penalty = 0.00499 as reference. Experimental rankings were generated with a higher range (0.05, 0.0499) and with a lower penalty (0.005, 0.002). These rankings yield Kendall's (tau) correlation coefficients of 0.918 and 0.934, respectively.

Table 10 Rank positions for alternative betweenness centrality measures

	Country	OCCUR	STRCT	UNWTD
1	UK	4	1	1
2	Luxembourg	2	2	10
3	Netherlands	8	4	2
4	Estonia	3	5	7
5	Hungary	1	7	5
6	Singapore	7	3	4
7	Ireland	5	6	14
8	Slovak Republic	6	8	11
9	Finland	11	12	28
10	Cyprus	17	10	3

would coincide; this is, however, not the case at all as explained above. In addition, an *Unweighted* (UNWTD) betweenness measure for relevant paths within range is created.

The main variant which we present is *Betweenness*. Its weighting and the range, with a cost for conduit entities, make it the most realistic measure. Besides we use the three extra measures to test whether the assumptions of BTWNS affect the outcomes. Table 10 presents the positions of the main top 10 in the alternative centrality rankings. In Appendix B3 the full list can be found.

The UK heads three of the four variants of the centrality measures. Hungary ranks first in the alternative measure of occurrence. This ranking, however, is not very different from the reference ranking of betweenness. The ranking with the strict optimal paths also does not differ much from that with a range. The same applies for the unweighted measure, which has the UK, the Netherlands, and Cyprus as top 3. Thus, with respect to the ranking the results of the centrality measures are robust.[27]

7 The impact of treaty shopping and centrality on FDI stocks

This section presents the effects of conduit countries and treaty shopping on bilateral FDI stocks using the indicators for the treaty shopping gain and centrality developed above. The results confirm the relevance of our indicators.[28]

[27] Kendall's (tau) rank correlation coefficients with BTWNS are 0.85 for OCCUR, 0.83 for STRCT and 0.82 for UNWTD.

[28] This analysis builds upon a huge empirical literature on macro-FDI and taxation. See for some surveys De Mooij and Ederveen (2003) and Feld and Heckemeyer (2011). Besides there are various papers estimating the effect of bilateral tax treaties on FDI (Blonigen and Davies 2004, 2005; Egger et al. 2006; Sauvant and Sachs 2009; Blonigen et al. 2014).

We conduct regressions for the 108 countries in our network, using the IMF/CDIS data for 2013 which include FDI via SPEs.[29] SPEs are likely vehicles for FDI diversion (OECD 2013). The CDIS database includes the bilateral FDI positions of about 100 countries reporting inward stocks and 80 countries reporting outward stocks. We start with the inward stocks and add outward stocks if the corresponding inward stocks are missing. We end up with 6422 observations of the possible 11,556 ones. The other observations are missing, by either non-reporting or reported "confidential data." It is likely that the missing values represent mainly smaller FDI stock values. Egger et al. (2009) find that the missing values in their sample are not randomly distributed using a Heckman selection model. As robustness check we run the regressions in a smaller sample with relatively fewer missing observations. This does not alter our main conclusions.

We explain the variation in bilateral FDI stocks (*FDI)* in a gravity model, using GDP in the host and home country and the quality of institutions in both countries as basic explanatory variables.[30] These data come from the World Bank Development Indicators, sometimes added with GDP data of tax havens from the CIA factbook. We add geographic variables from CEPII; distance, dummies for a common language, former colony, contiguity, former common country, both OECD countries, regional trade agreement, and a common legal system. CEPII distinguishes five legal systems: French, British, socialist, German, and Scandinavian. Except for distance, we expect positive coefficients for these dummies.

All these variables are part of the vectors G in Eq. (1). The subscripts refer to the home country, host country, and a bilateral relation. The vectors T refer to the tax variables.

$$\begin{aligned} FDI_{ij} &= \alpha_0 + \alpha_i G_i + \alpha_j G_j + \alpha_{ij} G_{ij} + \beta_i T_i + \beta_j T_j \\ &\quad + \beta_{ij} T_{ij} + \gamma_{ij} GAIN_{ij} + \gamma_i CENT_i + \gamma_j CENT_j + \varepsilon_{ij} \end{aligned} \tag{1}$$

We add dummy variables for tax havens (*TAXH*), for which we expect a positive coefficient. These dummies are based on the Gravelle (2013) list discussed in Sect. 3. We include the CIT rates of both countries (*CIT*), the standard withholding taxes on royalties (WHT^{roy}), the bilateral withholding taxes (*BWHT*) in tax treaties, and the remaining double taxation in the home country ($RCIT_{ij}$). The tax data come from the sources mentioned in Sect. 3.

The variable $GAIN_{ij}$ reflects the gains of lower taxation by using indirect routes for FDI. If the direct route between countries i and j is the cheapest one, $GAIN_{ij}$ is equal to zero. If there are cheaper options, $GAIN_{ij}$ has a positive value. We take the values of *GAIN* from the network analysis in Sect. 5. We expect a negative sign for the coefficient. *CENT* reflects the centrality indicator. In our main specification we

[29] See the definition of SPEs in Sect. 1.

[30] We have used the average of the six governance indicators for this purpose.

use the results from the betweenness indicator, Sect. 6. ε_{ij} represents the error terms which are assumed to be independent and identically distributed.[31]

A well-known problem in estimating gravity equations is the number of zero values for the dependent variable. This problem is extensively discussed in the trade literature (Head and Mayer 2014) with the many zeros for bilateral trade values in samples with developing countries. The zero values create missing values with a standard log transformation. This also applies for bilateral FDI stocks. The most common solution is a transformation of $\log(1+x)$ in which x is the bilateral trade value. This does not solve all problems. Because trade values cannot be negative, the error distribution is truncated and could depend on the explanatory variables. Santos-Silva and Terenyro (2006) argue for that reason to use a Poisson pseudo-maximum likelihood (PPML) method. The problem of our data sample is the extremely skewed distribution of the FDI values ranging from many zero values to values of more than 100 billion USD. About 1% of the observations exceeds 100 billion USD, and these cover 50% of the value of the aggregated bilateral FDI stocks in the sample.[32] To correct for the skewed distribution, we use the log transformation in an OLS estimation and present the results of the PPML estimator as a robustness analysis.

Another issue is the potential endogeneity of various explanatory variables with bilateral FDI stocks. The literature on the impact of bilateral tax treaties on FDI treats these treaties often as endogenous[33]; increased investment could raise the necessity of a treaty. Many conduit countries indeed signed many treaties. Moreover, the withholding tax rates could be endogenous. Chisik and Davies (2004) show theoretically and empirically that larger asymmetric FDI between countries implies higher tax rates. However, since the network centrality of a conduit country is measured as an average over all its 107 partner countries, the impact of a specific bilateral treaty and its negotiated tax rates on the network centrality of a country is only limited. This suggests that the potential endogeneity between network centrality and bilateral FDI is very indirect and small and therefore of minor importance in this study.

Table 11 presents the summary statistics.[34] FDI stocks and GDP are measured in logs. The corporate income tax rate varies from zero to 46% and is on average 23%. Of the 108 countries 23% is defined as a tax haven. The default withholding tax on royalties varies between 0 and 34% with an average of 15%. The bilateral withholding tax rates on dividends are on average 9%. The remaining CIT paid in the home country is on average 7%, but varies between nothing and 40%. The bilateral direct repatriation tax rate is 15% in this sample and is somewhat larger than the double GDP-weighted average tax rate in Sect. 5 for all country pairs. Quite often the direct repatriation tax

[31] Our empirical specification resembles Hong (2017). He also uses an OLS estimator in a cross section with the same geographic variables. We add the level of institutional quality and add more tax variables, such as the withholding tax rates on royalties and the bilateral withholding tax rate on dividends. Moreover, we combine the bilateral gains from treaty shopping and the centrality indicators in one specification. Another difference is our choice of using 2013 IMF/CDIS data, inclusive of SPEs, whereas Hong (2017) primarily uses 2012 UNCTAD data.

[32] Many of these bilateral FDI stocks have the USA, Luxembourg, or the Netherlands as resident or host country. This corresponds to the ordering of aggregated inward and outward FDI stocks in Table 1.

[33] Some references are Blonigen and Davies (2004, 2005) and Egger et al. (2006).

[34] Statistics and results for the non-tax variables, other than GDP, are presented in Appendix D1.

Table 11 Summary statistics

Variable	Mean	Std. dev.	Min	Max
Log FDI	3.31	3.43	0.00	13.92
Tax variables				
CIT rate	0.23	0.09	0.00	0.46
Tax haven dummy	0.23	0.42	0.00	1.00
Withholding tax on royalties (WHT^{roy})	0.15	0.10	0.00	0.34
Bilateral withholding tax (*BWHT*)	0.09	0.09	0.00	0.35
Remaining CIT (*RCIT*)	0.07	0.11	0.00	0.40
Direct repatriation tax rate (*DRCT*)	0.15	0.13	0.00	0.58
Treaty shopping indicators				
Optimal repatriation tax rate (*OPTM*)	0.05	0.08	0.00	0.47
Gain treaty shopping (*GAIN*)	0.10	0.11	0.00	0.50
Betweenness indicator	1.61	2.48	0.00	12.79
Occurrence indicator	27.73	25.44	0.00	76.90
Strict indicator	0.74	1.78	0.00	13.82
Unweighted indicator	1.73	2.48	0.00	12.79

Number of observations is 6422. The summary statistics of the gravity variables are presented in Appendix D

rate is zero, e.g. between the EU member states, but it could also be 58%, suggesting no double tax relief at all. In about two-thirds of the cases, it is optimal to divert FDI via other countries. This reduces the average bilateral tax rate to 5%. The average unweighted gain of treaty shopping is 10% points.

7.1 Regression results

The outcomes of the OLS regressions are found in Table 12. The standard errors are corrected for heteroscedasticity using the Huber–White estimator. Tax havens have larger inward and outward FDI stocks; the coefficients are positive and significant in both home and host countries. The coefficient of about 1 suggests that bilateral FDI stocks are twice as large if the home or host country is a tax haven. Tax havens distinguish themselves from other countries, not only because of low or negligible tax rates on profits, but also because of characteristics such as a lack of transparency, bank secrecy, lack of information sharing, and requiring little or no economic activity for an entity to obtain legal status (Gravelle 2013). These characteristics suggest that tax haven dummies and corporate tax rates are not strongly correlated. This proves also to be the case. CIT rates in the home country have a negative impact on the FDI stocks. The coefficient for the CIT rate of the host country is also significant. The semi-elasticity is about 3, suggesting an increase in the corporate tax rate of 1% points lowers the FDI stock by about 3%.

The default withholding taxes on royalties have a negative effect on bilateral FDI stocks for both countries. We have added these variables in column (2) because also tax treatment of royalty and interest payments affect FDI. Because of the multicollinearity

Table 12 Explaining FDI stocks with indicators for treaty shopping and conduit countries

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dummy TAXH_home	1.537***	1.364***	1.353***	1.334***	1.350***	1.340***	1.545***
	(0.0840)	(0.0849)	(0.0844)	(0.0842)	(0.0849)	(0.0844)	(0.0832)
Dummy TAXH_host	1.069***	0.867***	0.879***	0.952***	0.869***	0.880***	0.944***
	(0.0872)	(0.0888)	(0.0888)	(0.0878)	(0.0887)	(0.0887)	(0.0884)
CIT_home	−1.863***						
	(0.430)						
CIT_host	−3.066***	−2.347***	−2.506***	−3.016***	−2.283***	−2.437***	−2.619***
	(0.432)	(0.445)	(0.452)	(0.439)	(0.450)	(0.457)	(0.455)
WHT^{roy}_home		−3.849***	−3.790***	−3.720***	−3.743***	−3.685***	
		(0.332)	(0.334)	(0.334)	(0.336)	(0.338)	
WHT^{roy}_host		−0.765**	−0.750**	−1.073***	−0.669*	−0.657*	
		(0.370)	(0.370)	(0.365)	(0.366)	(0.366)	
BWHT		−3.383***	−2.612***		−3.029***	−2.299***	−2.160***
		(0.376)	(0.492)		(0.386)	(0.501)	(0.495)
RCIT		−1.006***	−0.599*		−0.942***	−0.553	−0.822**
		(0.296)	(0.337)		(0.297)	(0.338)	(0.339)
GAIN Treaty shopping			−0.993**	−2.548***		−0.948**	−1.351***
			(0.433)	(0.305)		(0.433)	(0.433)
CENT_home					0.0297**	0.0299**	0.0569***
					(0.0149)	(0.0149)	(0.0148)
CENT_host					0.0541***	0.0530***	0.0557***
					(0.0172)	(0.0172)	(0.0175)
Constant	−6.874***	−7.379***	−7.358***	−7.315***	−7.654***	−7.631***	−7.262***
	(0.450)	(0.441)	(0.440)	(0.440)	(0.450)	(0.449)	(0.456)
Adjusted R^2	0.528	0.545	0.545	0.543	0.546	0.546	0.538

The regressions are estimated with OLS and Huber–White robust standard errors. There are 6422 observations. Robust standard errors are in parentheses, ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. The coefficients for the gravity variables are presented in Appendix D

of both variables, we have only included the withholding tax rates on royalties. The semi-elasticities are about 3.8 and 0.8 for the home and host country, respectively. We also include the bilateral withholding taxes (*BWHT*) agreed upon in tax treaties. We find a negative significant effect; the semi-elasticity is 3.4, suggesting that 1%-point lower tax rates increase the bilateral FDI stock by nearly 3.4%. A 1%-point lower bilateral remaining CIT rate (*RCIT*) in the home country raises bilateral FDI by 1%.

In column (3) of Table 12 we add the treaty shopping gain indicator (*GAIN*). If the direct route is not the optimal one, the bilateral FDI stock is lower; the coefficient is statistically significant. If the repatriation tax is lowered by 1% point due to treaty shopping ($GAIN > 0$), the bilateral FDI stock decreases by 1%. The coefficients of the bilateral withholding tax on dividends and the remaining CIT rate are lower now, which is probably due to correlation with treaty shopping. The correlation between *BWHT* and *GAIN* is 0.63 and between *RCIT* and *GAIN* 0.47. There are two reasons for this result. First, in many bilateral relations both tax rates are zero: no withholding tax in the host country and exemption in the home country. By definition, then there is no cheaper tax route, so the treaty shopping variable is zero. Second, the higher the withholding tax rate and the remaining CIT rate are, the higher is the probability that there is an optimal route with lower tax rates. If we ignore both taxes in the regression the coefficient of the treaty shopping gain is much larger as can been seen in column (4). However, in this case the coefficient also includes the beneficial investment effects of a zero withholding tax rate and exemption. It is one of our contributions that we want to distinguish these effects. Therefore, we include all variables in the remaining specifications.

Next, we switch to the role of conduit countries in column (4). Our preferred network centrality indicator is *Betweenness*. A high value of this indicator implies that the country is often used as a conduit country for diverting FDI and profit flows, because the value reflects the share of tax-minimizing routes on which the country is present as conduit. We expect that a conduit country receives much FDI and diverts much FDI to other countries. These could be other conduit countries or final host countries. The centrality indicators in the host country and in the home country have a positive and significant impact on FDI. An increase of one standard deviation of the centrality indicator raises the bilateral FDI stock by about 10% in the home country and 20% in the host country. This confirms our intuition; if the home or host country of the bilateral FDI stock is a conduit, FDI stocks are larger.

Although a high value of the centrality indicator suggests that a country is used for treaty shopping, we also include $GAIN_{ij}$ in the regression to evaluate the impact of treaty shopping between both countries on bilateral FDI. When a particular direct investment route is not optimal, in spite of a conduit country on this direct route, treaty shopping has still a negative effect on the bilateral FDI. This is confirmed with the regression shown in column (5).

Conduit countries are not only attractive because of exemption as double tax relief method and low taxes on dividends, but also because of low withholding taxes on royalties and interest. The negative correlation between the withholding taxes on royalties and the centrality indicator is large. Therefore, we exclude the withholding taxes on royalties from the regression. Then the coefficient for the centrality indicator of the home country is twice as large, suggesting indeed multicollinearity.

7.2 Robustness analysis

The coefficients of the centrality indicators for the home and host country have a positive and significant impact on bilateral FDI stocks. Moreover, we also find that bilateral FDI stocks are larger if the direct profit repatriation route is also the optimal tax route. We verify that these conclusions also hold for other specifications.

First, we estimate the impact of the treaty shopping gain with fixed effects for the home and host countries. This excludes the possibility of omitted country-specific variables, but also throws the baby out with the bathwater. The centrality indicators are our main interest. They are country specific and therefore "included" in the fixed effects in this specification. The coefficient for the treaty shopping gain is still negative and significant at the 90% level. Due to the correlation with the other bilateral tax variables discussed above, the standard errors are quite large now. As an alternative we have replaced the treaty shopping gain by a dummy, which is one if the direct route is the optimal one and zero if it is not. Hong (2017) also uses this dummy. This reduces the correlation and the coefficient for *BWHT* becomes significant. The coefficient for the dummy is positive and significant.[35]

Second, we apply the PPML estimator instead of OLS. The dependent variable is FDI and includes all zero values. Now, the variance of the dependent variable is much larger, because the log transformation reduces the skewed distribution of the FDI values ranging from many zero values to values below 100 billion USD. About 1% of the observations exceeds 100 billion USD and covers 50% of the value of the aggregated bilateral FDI stocks in sample. The coefficients of treaty shopping gain and the centrality indicators are still significant and have the expected sign. Neither the coefficient for the CIT rate in the host country nor the coefficient for the bilateral withholding tax is significant now. The coefficient of the remaining CIT rate in the home country is positive. The log transformation reduces the impact of the large FDI stocks of the US, which are positively correlated with the high effective tax rates in the US. With PPML, we find a large correlation between bilateral FDI and these high effective tax rates. The effects of the treaty shopping gain on FDI are larger, and the centrality indicator in the home country also has more impact on FDI (Table 13).

Many empirical papers on FDI stocks focus only on the OECD countries as home and host countries of foreign investment; see Egger et al. (2009) and Benassy-Quere et al. (2007) among others. These are the most important outward investing countries. An advantage is that the data selection contains fewer missing observations and fewer zero values for bilateral stocks.[36] This is also a check to see whether the large share of missing and zero values in the full sample affects the regression results. We restrict our country sample for the home countries to 34 OECD countries and all 108 countries as host countries for investment. The coefficients of the centrality indicators are larger,

[35] The results are available upon request. We have also eliminated *BWHT* and *RCIT*. Then the coefficient for *GAIN* is -2.721 and statistically significant.

[36] After correcting for missing values for GDP, institutions, and geographic variables, we have 10609 observations in the full sample. 4187 observations do not have a value for the bilateral FDI stock (about 40%). 1902 observations have a zero value for the FDI stock (18%). In the smaller OECD sample we miss 719 observations of the possible 3502 observations which is 20%. The number of zero values for FDI is 484, 14% of the total.

Table 13 Robustness analyses

Robustness	(1) Fixed effects	(2) PPML	(3) OECD sample	(4) Tax haven	(5) Occurrence	(6) Strict	(7) Unweighted
Dummy TAXH_home		0.988***	1.543***	1.457***	1.351***	1.326***	1.278***
		(0.165)	(0.166)	(0.0965)	(0.0849)	(0.0844)	(0.0838)
Dummy TAXH_host		0.642***	0.713***	0.835***	0.939***	0.859***	0.824***
		(0.194)	(0.132)	(0.0977)	(0.0902)	(0.0887)	(0.0883)
CIT_host		0.207	−2.518***	−2.548***	−2.059***	−2.494***	−2.239***
		(1.329)	(0.649)	(0.463)	(0.477)	(0.453)	(0.457)
WHT^{roy}_home		−1.308**	−2.785***	−4.033***	−3.693***	−3.664***	−3.275***
		(0.574)	(0.462)	(0.330)	(0.336)	(0.335)	(0.335)
WHT^{roy}_host		−3.188***	0.0976	−0.900**	−0.749**	−0.707*	−0.509
		(0.710)	(0.482)	(0.364)	(0.369)	(0.368)	(0.362)
BWHT	−0.872	−1.948	−1.705*	−2.153***	−1.863***	−2.442***	−1.559***
	(1.833)	(1.428)	(0.875)	(0.500)	(0.515)	(0.496)	(0.505)
RCIT	2.314	4.071***	0.143	−0.499	−0.354	−0.542	−0.220
	(1.769)	(1.134)	(0.722)	(0.339)	(0.347)	(0.337)	(0.337)
GAIN Treaty shopping	−3.558*	−3.647**	−1.257	−1.034**	−1.179***	−0.925**	−1.100**
	(1.943)	(1.514)	(0.785)	(0.434)	(0.436)	(0.433)	(0.431)
CENT_home		0.0798***	0.0904***	0.0704***	0.00351**	0.0669***	0.0987***
		(0.0218)	(0.0162)	(0.0148)	(0.00155)	(0.0165)	(0.0140)
CENT_host		0.0629***	0.0722***	0.0784***	0.00755***	0.0525**	0.105***
		(0.0241)	(0.0208)	(0.0176)	(0.00176)	(0.0206)	(0.0167)
Constant	8.889***	10.66***	−15.34***	−7.862***	−8.211***	−7.453***	−7.997***
	(0.651)	(1.365)	(0.756)	(0.453)	(0.481)	(0.440)	(0.444)
Adjusted R^2	0.732	0.434	0.638	0.544	0.547	0.547	0.552

The regressions are estimated with OLS, except for column (2). There are 6422 observations, except in the OECD sample where we have 2783 observations. Robust standard errors are in parentheses, ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. The coefficients for the gravity variables are presented in Appendix D

but the coefficient on treaty shopping is not significant now. The coefficient on the bilateral tax rate is hardly significant. This could be due to a lack of variation of these variables. The sample is much smaller and a large share of the country pairs is EU countries. Because of the Parent–Subsidiary directive, the bilateral tax rates and the gains of treaty shopping are zero.

Column (4) presents another definition of the tax haven variable. The definition of a tax haven comprised of many conduit countries, which we try to represent with the centrality indicator. So we exclude the financial centres Ireland, Jordan, Luxembourg, Singapore, and Switzerland as tax havens. Three of these countries are in the top 10 of countries with highest inward FDI stocks (Table 1). The quantitative impact of centrality on FDI stocks is somewhat larger now, as is the impact of the tax haven dummy on outward FDI stocks. All other results are comparable suggesting that the precise definition of tax haven does not matter for our conclusions.

Section 6 defines other centrality indicators: *Strict*, *Occurrence*, and *Unweighted*.[37] With these other centrality indicators, the coefficients for centrality are again positive and significant, except for occurence in the home country. Columns (5) to (7) show that the impact of a conduit country on the bilateral FDI stocks is robust. For all indicators we find the same pattern. The coefficient for treaty shopping gain is also negative and significant.

8 Conclusion

We have presented a novel perspective by applying a network analysis to international corporate taxation. This yields the contribution of FDI diversion, and the corresponding dividend flows, to the reduction of the tax burden of multinational enterprises. At the same time it provides insight into the central position of particular countries in the international tax network. We have modelled corporate taxation in host and home countries, double tax relief methods, and the withholding tax on dividends for all pairs in a sample of 108 countries. The direct effect of double tax relief and tax treaties lowers the world average repatriation tax rate from a theoretical 41–12%.

Our first result is that the possibility of treaty shopping allows for a further reduction of 6% points, leaving, on average, a double taxation on repatriated dividends of 6%. For about two-thirds of the country pairs examined, there exists an indirect tax route that is more attractive in terms of lower taxes than the direct route. A large cluster of 68 countries exists that are well interconnected through zero tax routes. These results imply that it can be very beneficial for multinational companies to restructure their investments over various countries in such a way that tax payments are minimized.

Centrality in the network is used to identify candidates for the role of conduit country. This leads to the second result. The UK heads the ranking of network centrality, followed by Luxembourg and the Netherlands. The top 10 has only one non-EU country: Singapore. Ireland is the third tax haven financial centre in the top 10. In contrast,

[37] We use in the main regressions the network centrality indicator *Betweenness* which is double GDP-weighted. The weighting involves all bilateral flows except those with the country concerned as a starting or endpoint. Therefore, own GDP is not part of the measure and there is no endogeneity problem.

low-tax havens such as Bermuda, the Cayman Islands and the British Virgin Islands are not crucial conduit countries for the treaty shopping motive. The centrality results are robust with respect to alternative measures of centrality.

The third result concerns the empirical relevance of our indicators. We regress bilateral FDI stocks on the outputs of the network analysis, the treaty shopping gain indicator, and the (unilateral) network centrality measures for both host and home country. These explanatory variables are found to be significant, statistically and in terms of economic impact. An increase of one standard deviation of the centrality indicator raises the bilateral FDI stock by about 10% in the home country and 20% in the host country. If firms can reduce bilateral taxation by 1% point by treaty shopping, the bilateral FDI stock decreases by 1%. Moreover, the indicators are significant in an array of robustness checks.

This brings us to the limitations of the study and directions for further research. We take the profits in the host country as given and focus on dividend flows. More generally, a broader fiscal and juridical environment will affect the holding decisions of multinationals and the size of taxable profit incomes. These activities may involve intra-company financing and the location of intellectual property rights so that deductibility of interest and royalty payments matter and the withholding taxes for these categories. We ignore the possibilities to reduce the tax base with interest and royalty payments. Next, our analysis lacks dynamics and we require profits to be repatriated to the home country. Thus, deferral is no option, and we miss out on the parking function associated with traditional low-tax havens, as discussed by Mintz and Weichenrieder (2010). Finally, the econometric analysis could go beyond cross-sectional regressions. The impact of the centrality measure on FDI stocks could be analysed in a panel setting with country-pair fixed effects, although the time variation with respect to the tax parameters is limited. Another possibility would be to investigate the relationship between FDI stocks and flows in more depth using spatial regression analysis as in Blonigen et al. (2007). Even without FDI diversion these authors argue that FDI stocks in various regions could be interdependent. For FDI stocks via conduit countries, the interdependence of stocks between various regions could be even stronger.

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Appendix A1: Collected tax data 2013—108 jurisdictions

Tax variable (unit)	CIT rate (%) (1)	Relief method (2)	Treaty relief method (3)	CFC rule (dummy) (4)	WHT^{div} (%) (5)	DTT (no.) (6)	Tax haven (dummy) (7)	GDP weight (% point) (8)
Albania	10	3	3	0	10	26	0	0.03
Algeria	25	0	0	0	15	23	0	0.34
Angola	35	0	0	0	10	0	0	0.16
Argentina	35	3	3	0	35	14	0	0.94
Aruba	28	4	4	0	10	1	1	0.00
Australia	30	3	3	1	30	40	0	1.23
Austria	25	4	4	0	25	66	0	0.45
Azerbaijan	20	3	3	0	10	29	0	0.12
Bahamas	0	4	4	0	0	0	1	0.01
Bahrain	46	3	3	0	0	10	1	0.04
Barbados	25	3	3	0	15	23	1	0.01
Belarus	18	3	3	0	12	44	0	0.19
Belgium	34	5	5	0	25	70	0	0.53
Bermuda	0	4	4	0	0	0	1	0.01
Botswana	22	2	2	0	7.5	8	0	0.04
Brazil	34	3	3	1	15	35	0	2.97
Brunei Darus-salam	20	4	4	0	0	1	0	0.03
Bulgaria	10	3	3	0	5	50	0	0.13
Canada	26.3	3	4	0	25	75	0	1.88
Cayman Islands	0	4	4	0	0	0	1	0.00
Chile	20	2	3	0	35	24	0	0.40
China	25	3	3	1	10	61	0	15.66
Colombia	25	3	3	0	0	4	0	0.63
Costa Rica	30	2	3	0	15	1	1	0.07
Croatia	20	3	3	0	12	44	0	0.10
Curacao	27.5	4	4	0	0	0	0	0.00
Cyprus	12.5	4	4	0	0	35	1	0.03
Czech Republic	19	2	3	0	35	66	0	0.36
Denmark	25	3	4	1	27	61	0	0.27
Dominican Rep.	29	3	3	0	10	1	0	0.12
Ecuador	22	0	0	0	0	10	0	0.19

Tax variable (unit)	CIT rate (%) (1)	Relief method (2)	Treaty relief method (3)	CFC rule (dummy) (4)	WHT^{div} (%) (5)	DTT (no.) (6)	Tax haven (dummy) (7)	GDP weight (% point) (8)
Egypt	25	2	3	0	0	23	0	0.68
Equatorial Guinea	35	0	0	0	25	1	0	0.02
Estonia	21	3	4	0	0	36	0	0.04
Finland	24.5	3	4	1	24.5	59	0	0.25
France	34.3	5	5	1	30	80	0	2.84
Gabon	35	2	3	0	15	4	0	0.03
Germany	30.2	5	5	1	25	71	0	4.04
Greece	26	3	3	0	10	42	0	0.35
Guernsey	0	4	4	0	0	0	1	0.00
Hong Kong	16.5	4	4	0	0	14	1	0.47
Hungary	19	4	4	1	0	47	0	0.25
Iceland	20	3	3	0	18	38	0	0.02
India	34	3	3	0	0	40	0	5.91
Indonesia	25	3	3	0	20	52	0	1.54
Ireland	12.5	3	3	0	20	53	1	0.24
Isle of Man	0	4	4	0	0	0	1	0.01
Israel	25	3	3	1	20	43	0	0.31
Italy	31.4	5	5	1	20	69	0	2.31
Jamaica	25	2	3	0	33.33	15	0	0.03
Japan	37	5	5	0	20	47	0	5.84
Jersey	0	4	4	0	0	0	1	0.01
Jordan	14	2	3	0	0	13	1	0.05
Kazakhstan	20	3	3	0	15	35	0	0.29
Korea Republic	24.2	3	3	0	20	67	0	2.04
Kuwait	15	2	2	0	15	40	0	0.19
Latvia	15	3	3	0	10	45	0	0.05
Lebanon	15	2	3	0	10	13	1	0.08
Libya	20	0	3	0	0	1	0	0.10
Liechtenstein	12.5	4	4	0	0	3	1	0.00
Lithuania	15	3	4	0	15	44	0	0.08
Luxembourg	29.2	4	4	0	15	57	1	0.05
Macao	12	0	0	0	0	0	1	0.06
Malaysia	25	4	4	0	0	34	0	0.63
Malta	35	4	4	0	0	38	1	0.01
Mauritius	15	3	3	0	0	15	1	0.03
Mexico	30	2	2	0	0	36	0	2.22

Tax variable (unit)	CIT rate (%)	Relief method	Treaty relief method	CFC rule (dummy)	WHT^{div} (%)	DTT (no.)	Tax haven (dummy)	GDP weight (% point)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mongolia	25	2	3	0	20	27	0	0.02
Namibia	34	3	3	0	10	9	0	0.02
Netherlands	25	4	4	0	15	74	0	0.89
New Zealand	28	3	3	1	30	36	0	0.17
Nigeria	30	3	3	0	10	11	0	0.57
Norway	28	3	3	1	25	64	0	0.35
Oman	12	3	3	0	0	8	0	0.11
Pakistan	35	3	3	0	10	31	0	0.65
Panama	25	0	0	0	17	14	1	0.07
Peru	30	3	3	1	4.1	3	0	0.41
Philippines	30	3	3	0	15	29	0	0.54
Poland	19	3	4	0	19	64	0	1.01
Portugal	31.5	3	3	0	25	53	0	0.31
Puerto Rico	30	3	3	0	10	0	0	0.08
Qatar	10	2	3	0	7	36	0	0.24
Romania	16	3	3	0	16	66	0	0.35
Russian Federation	20	3	3	0	15	59	0	3.17
Saudi Arabia	20	0	3	0	5	18	0	1.14
Serbia and Mont.	15	3	3	0	20	42	0	0.10
Seychelles	33	0	0	0	15	12	1	0.00
Singapore	17	4	4	0	0	40	1	0.41
Slovak Republic	23	2	3	0	0	42	0	0.17
Slovenia	17	3	3	0	15	46	0	0.07
South Africa	28	3	3	1	15	55	0	0.74
Spain	30	4	4	1	21	71	0	1.78
Suriname	36	3	3	0	25	1	0	0.01
Sweden	22	3	3	1	30	67	0	0.50
Switzerland	21.1	4	4	0	35	71	1	0.46
Taiwan Province	17	3	3	1	20	19	0	1.14
Thailand	20	2	3	0	10	34	0	0.82
Trinidad and Tob.	25	3	3	0	10	16	0	0.03
Tunisia	30	0	0	0	0	26	0	0.13
Turkey	20	3	3	1	15	59	0	1.42

Tax variable (unit)	CIT rate (%) (1)	Relief method (2)	Treaty relief method (3)	CFC rule (dummy) (4)	WHT^{div} (%) (5)	DTT (no.) (6)	Tax haven (dummy) (7)	GDP weight (% point) (8)
Ukraine	19	3	3	0	15	56	0	0.42
United Arab Emirates	0	4	4	0	0	21	0	0.34
UK	23	4	4	0	0	51	0	2.95
USA	39.1	3	3	1	30	54	0	19.79
Uruguay	25	2	2	0	7	6	0	0.07
Venezuela	34	3	3	0	34	28	0	0.51
Virgin Islands USA	38.5	3	3	0	11	0	1	0.00
Virgin Islands UK	0	4	4	0	0	0	1	0.00

Columns (2) and (3): 0 = no relief, 2 = deduction, 3 = credit, 4 = exemption, 5 = exemption 95%
Columns (4) and (7) are dummy variables: 1 = yes, 0 = no

Appendix A2: Tax havens

The first three columns give the intersection of the Gravelle (2013) list of 50 tax havens with our selection of 108 jurisdictions. The first column presents the low-tax havens, i.e. those with a corporate income tax rate in 2013 of 12.5% or less. Most of them are islands. Malta could have been included in the list of low-tax havens because the effective rate may be reduced to 10% or even less by a refund mechanism, although it has a nominal rate of 35% (Loyens and Loeff 2013). We refrain from using such characteristics and stick to the bare tax parameters (Table 14).

The second column indicates the five countries from the Gravelle list that we have treated differently with respect to the CFC rules. We believe these five countries are not subjected to the same anti-abuse scrutiny as are the jurisdictions in the first and third column. The Gravelle list is based on an overview of other papers classifying tax havens. The first four appear only in the list of Dharmapala and Hines (2009) and Hines and Rice (1994), and Singapore is often considered as another financial centre, different from other tax havens (OECD "grey" list).

The third column contains the other tax havens in our analysis that are deprived of preferential tax relief by those countries indicated to apply CFC rules. These are neither low-tax havens nor financial centres. The fourth column contains, for comparison, the other countries in the country set with a low tax rate, i.e. a CIT of 12.5% or less.

Table 14 Tax haven lists

Low-tax havens	Tax haven financial centres	Other tax havens	Other low-tax countries
Bahamas	Ireland	Aruba	Albania
Bermuda	Jordan	Bahrain	Bulgaria
Cayman Islands	Luxembourg	Barbados	Oman
Cyprus	Singapore	Costa Rica	Qatar
Guernsey	Switzerland	Hong Kong	United Arab Emirates
Isle of Man		Lebanon	
Jersey		Malta	
Liechtenstein		Mauritius	
Macao		Panama	
Virgin Islands UK		Seychelles	
		Virgin Islands USA	

Appendix B1: Average dividend repatriation for home countries

Tax variables (unit)	Avg. BWHT inward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Albania	13.72	13.76	1.64	12.12	29
Algeria	13.35	35.01	26.23	8.78	103
Angola	17.39	46.31	36.06	10.25	107
Argentina	14.76	17.83	8.19	9.65	90
Aruba	17.38	17.38	1.64	15.74	12
Australia	6.38	8.01	1.77	6.24	78
Austria	4.82	4.82	1.65	3.17	58
Azerbaijan	13.93	14.08	1.63	12.45	10
Bahamas	17.39	17.39	1.64	15.75	22
Bahrain	15.35	27.11	23.26	3.86	98
Barbados	9.87	10.27	1.64	8.63	21
Belarus	9.96	10.09	1.64	8.45	43
Belgium	4.28	5.56	1.65	3.91	62
Bermuda	17.38	17.38	1.64	15.75	14
Botswana	16.56	34.92	23.28	11.64	99
Brazil	15.08	17.84	7.20	10.64	88
Brunei Darussalam	16.60	16.60	1.64	14.97	26
Bulgaria	6.69	6.72	1.64	5.08	40
Canada	6.25	6.40	1.57	4.83	5

Tax variables (unit)	Avg. BWHT inward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Cayman Islands	17.38	17.38	1.64	15.74	15
Chile	14.71	26.60	1.62	24.98	7
China	7.74	8.59	1.01	7.58	1
Colombia	16.44	16.88	1.65	15.23	64
Costa Rica	17.10	41.46	6.55	34.91	84
Croatia	11.74	11.94	1.64	10.30	38
Curacao	17.38	17.38	1.64	15.74	16
Cyprus	7.52	7.52	1.64	5.88	27
Czech Republic	5.41	6.70	1.64	5.06	55
Denmark	4.20	4.58	1.64	2.94	48
Dominican Rep.	17.26	18.12	1.63	16.50	8
Ecuador	15.27	33.91	23.28	10.63	100
Egypt	9.71	17.85	1.65	16.20	65
Equatorial Guinea	17.38	46.30	36.06	10.24	108
Estonia	6.56	6.72	1.64	5.08	30
Finland	3.38	3.65	1.64	2.01	46
France	3.29	4.91	1.69	3.23	74
Gabon	16.71	44.29	16.30	27.99	96
Germany	4.66	6.08	1.71	4.37	76
Greece	11.29	11.75	1.64	10.11	53
Guernsey	17.38	17.38	1.64	15.74	17
Hong Kong	13.80	13.80	1.65	12.15	60
Hungary	5.63	5.79	1.64	4.15	47
Iceland	6.42	6.56	1.64	4.92	23
India	10.56	13.57	7.56	6.01	89
Indonesia	9.23	9.64	1.66	7.97	71
Ireland	5.53	5.58	1.64	3.94	45
Isle of Man	17.38	17.38	1.64	15.75	18
Israel	9.54	10.17	1.64	8.53	50
Italy	5.56	7.00	1.68	5.32	73
Jamaica	10.23	20.30	1.64	18.66	31
Japan	5.99	7.73	3.56	4.17	82
Jersey	17.38	17.38	1.64	15.75	19
Jordan	15.42	25.30	6.56	18.74	85
Kazakhstan	7.51	7.67	1.63	6.04	9
Korea Republic	6.66	7.10	1.57	5.53	3
Kuwait	12.05	25.24	16.40	8.85	97
Latvia	6.24	6.31	1.64	4.67	33

Tax variables (unit)	Avg. BWHT inward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Lebanon	17.39	28.82	1.64	27.18	36
Libya	17.36	33.81	6.56	27.25	86
Liechtenstein	16.32	16.32	1.64	14.68	20
Lithuania	6.25	6.31	1.64	4.67	37
Luxembourg	4.03	4.03	1.64	2.39	34
Macao	17.39	27.31	13.44	13.86	94
Malaysia	11.81	11.81	1.65	10.16	63
Malta	6.59	6.59	1.64	4.95	24
Mauritius	14.78	14.86	1.64	13.22	28
Mexico	4.12	32.88	31.10	1.78	104
Mongolia	13.30	23.59	1.64	21.95	25
Namibia	15.55	18.10	6.99	11.11	87
Netherlands	3.39	3.39	1.65	1.74	67
New Zealand	8.46	9.46	1.64	7.82	41
Nigeria	16.06	17.07	1.78	15.30	79
Norway	7.01	7.77	1.64	6.12	54
Oman	16.11	16.16	1.64	14.52	39
Pakistan	10.68	13.89	8.27	5.63	91
Panama	15.56	36.67	26.23	10.44	101
Peru	16.61	17.76	1.79	15.97	80
Philippines	11.49	12.57	1.76	10.81	77
Poland	6.26	6.29	1.66	4.64	68
Portugal	8.51	9.66	1.64	8.01	49
Puerto Rico	17.39	18.38	1.80	16.58	81
Qatar	14.59	20.20	1.64	18.55	44
Romania	7.46	7.54	1.64	5.89	51
Russian Federation	6.49	6.68	1.53	5.15	2
Saudi Arabia	12.96	22.62	1.66	20.96	70
Serbia and Mont.	13.04	13.12	1.64	11.48	11
Seychelles	16.35	43.96	34.10	9.86	106
Singapore	10.84	10.84	1.65	9.19	56
Slovak Republic	5.60	13.05	1.64	11.41	42
Slovenia	6.42	6.51	1.64	4.87	35
South Africa	5.44	6.63	1.65	4.98	66
Spain	6.50	6.76	1.67	5.09	72
Suriname	17.38	20.69	9.77	10.91	92

Tax variables (unit)	Avg. BWHT inward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Sweden	3.48	3.83	1.65	2.18	61
Switzerland	4.92	4.92	1.65	3.27	59
Taiwan Province	14.95	15.18	1.66	13.52	69
Thailand	10.00	18.47	1.57	16.90	4
Trinidad and Tob.	9.56	9.96	1.64	8.32	32
Tunisia	11.44	38.01	31.15	6.86	105
Turkey	10.58	10.91	1.59	9.32	6
Ukraine	6.24	6.37	1.65	4.73	57
United Arab Emirates	13.19	13.19	1.64	11.54	52
UK	3.75	3.75	1.69	2.06	75
USA	6.29	16.67	14.61	2.06	95
Uruguay	16.05	37.04	26.23	10.81	102
Venezuela	7.95	10.86	6.49	4.37	83
Virgin Islands USA	17.38	22.18	13.23	8.95	93
Virgin Islands UK	17.38	17.38	1.64	15.74	13

Column (1) is the average withholding tax on dividends in the host countries repatriated to the home country Column (4) is the difference between columns (2) and (3). The results in both columns follow from the network analysis

The high rankings of China, Russia, and South Korea are explained by their relatively high outbound repatriation tax rates. Given the size of their economies, these rates contribute to higher inbound rates for the other countries and thus the three stand out with lower average inbound rates

Appendix B2: Average dividend repatriation tax rates for host countries

Tax variables (unit)	Avg. BWHT outward (%)	Min. BWHT outward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Albania	9.00	0	19.91	5.40	14.51	45
Algeria	11.67	0	13.99	5.34	8.66	22
Angola	10.00	10	11.72	10.88	0.83	98
Argentina	30.42	10	31.73	14.84	16.89	104
Aruba	9.96	5	28.53	10.04	18.48	89
Australia	10.77	0	15.14	4.69	10.45	13
Austria	6.79	0	11.26	5.42	5.84	70
Azerbaijan	9.58	5	15.29	10.13	5.16	92

Tax variables (unit)	Avg. BWHT outward (%)	Min. BWHT outward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Bahamas	0.00	0	26.37	5.40	20.98	39
Bahrain	0.00	0	20.57	4.89	15.67	16
Barbados	9.78	0	28.77	5.40	23.37	38
Belarus	10.05	0	15.60	5.41	10.20	57
Belgium	6.19	0	9.10	3.46	5.64	6
Bermuda	0.00	0	26.37	5.40	20.97	35
Botswana	7.29	5	13.29	10.12	3.16	90
Brazil	7.01	0	8.71	4.61	4.10	11
Brunei Darus-salam	0.00	0	11.21	5.40	5.81	43
Bulgaria	4.00	0	17.46	5.40	12.06	55
Canada	8.76	5	12.93	9.48	3.44	86
Cayman Islands	0.00	0	26.37	5.40	20.97	30
Chile	27.86	0	29.58	10.15	19.43	94
China	9.41	5	13.77	10.70	3.07	97
Colombia	0.00	0	7.70	5.43	2.28	75
Costa Rica	14.83	5	32.41	19.16	13.25	107
Croatia	8.63	0	13.75	5.40	8.34	53
Curacao	0.00	0	6.65	5.31	1.33	21
Cyprus	0.00	0	19.93	5.40	14.54	44
Czech Republic	7.50	0	14.37	5.42	8.95	67
Denmark	7.24	0	12.71	5.41	7.30	62
Dominican Rep.	10.00	10	12.72	11.89	0.83	100
Ecuador	0.00	0	9.67	5.37	4.30	23
Egypt	0.00	0	7.56	5.43	2.13	76
Equatorial Guinea	25.00	15	26.45	25.76	0.69	108
Estonia	0.00	0	10.07	5.40	4.67	48
Finland	5.87	0	11.69	5.41	6.28	61
France	5.23	0	8.07	3.41	4.65	4
Gabon	15.00	15	16.65	15.87	0.78	106
Germany	8.11	0	11.18	4.76	6.42	14
Greece	6.72	0	9.91	5.42	4.49	66
Guernsey	0.00	0	26.37	5.40	20.97	31
Hong Kong	0.00	0	23.37	5.42	17.95	72
Hungary	0.00	0	11.35	5.41	5.94	60
Iceland	7.66	0	14.06	5.40	8.66	40
India	0.00	0	3.86	2.97	0.88	3

Tax variables (unit)	Avg. BWHT outward (%)	Min. BWHT outward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Indonesia	11.05	0	14.54	5.48	9.06	81
Ireland	2.66	0	16.20	5.41	10.79	59
Isle of Man	0.00	0	26.37	5.40	20.97	32
Israel	12.24	0	14.76	5.41	9.35	63
Italy	8.04	0	10.60	4.33	6.27	9
Jamaica	20.06	0	23.36	5.40	17.96	46
Japan	7.25	0	9.24	2.48	6.77	2
Jersey	0.00	0	26.37	5.40	20.97	36
Jordan	0.00	0	14.96	5.40	9.57	33
Kazakhstan	8.56	5	15.04	10.14	4.90	93
Korea Republic	8.89	0	12.21	9.95	2.26	87
Kuwait	10.10	0	17.26	5.38	11.88	24
Latvia	5.62	0	15.35	5.40	9.95	49
Lebanon	9.47	0	29.30	5.40	23.90	51
Libya	0.00	0	11.19	5.40	5.80	37
Liechtenstein	0.00	0	24.15	5.40	18.75	34
Lithuania	6.64	0	16.10	5.40	10.69	52
Luxembourg	1.79	0	6.92	4.86	2.06	15
Macao	0.00	0	24.26	5.39	18.86	27
Malaysia	0.00	0	7.61	5.43	2.19	74
Malta	0.00	0	16.89	4.90	11.99	17
Mauritius	0.00	0	23.62	5.40	18.23	42
Mexico	0.00	0	4.86	4.05	0.81	7
Mongolia	13.73	0	15.14	5.40	9.74	41
Namibia	9.49	5	11.25	7.64	3.61	85
Netherlands	4.47	0	10.05	5.44	4.61	77
New Zealand	12.90	0	17.80	5.19	12.61	18
Nigeria	9.31	7.5	11.67	9.96	1.71	88
Norway	10.64	0	11.94	5.20	6.74	19
Oman	0.00	0	16.32	5.40	10.92	54
Pakistan	8.12	3.75	10.27	6.79	3.48	83
Panama	15.61	0	32.98	5.39	27.60	25
Peru	4.10	4.1	8.12	7.30	0.82	84
Philippines	13.04	10	14.52	12.60	1.92	102
Poland	7.80	0	13.75	5.45	8.31	78
Portugal	9.28	0	12.00	4.22	7.78	8
Puerto Rico	10.00	10	12.44	11.61	0.83	99
Qatar	6.13	0	19.15	5.41	13.74	58
Romania	8.74	0	14.83	5.42	9.41	65

Tax variables (unit)	Avg. BWHT outward (%)	Min. BWHT outward (%)	Direct routes (%)	Optimal routes (%)	Gain TS (%)	Rank optimal (no.)
Russian Federation	8.62	5	15.20	10.29	4.91	96
Saudi Arabia	4.78	0	12.71	5.46	7.26	80
Serbia and Mont.	13.04	5	18.75	10.13	8.62	91
Seychelles	13.00	0	30.82	5.40	25.43	28
Singapore	0.00	0	13.00	5.42	7.58	68
Slovak Republic	0.00	0	8.79	5.41	3.39	56
Slovenia	6.75	0	14.98	5.40	9.58	50
South Africa	6.68	0	10.42	5.22	5.20	20
Spain	7.84	0	9.88	4.69	5.19	12
Suriname	24.86	7.5	26.32	12.49	13.83	101
Sweden	5.97	0	12.50	5.42	7.08	73
Switzerland	7.31	0	13.03	5.42	7.61	71
Taiwan Province	17.71	0	21.29	5.46	15.83	79
Thailand	10.00	10	15.56	14.70	0.86	103
Trinidad and Tob.	8.95	0	12.95	5.40	7.55	47
Tunisia	0.00	0	5.39	4.56	0.83	10
Turkey	10.29	0	15.05	10.20	4.85	95
Ukraine	7.14	0	14.19	5.42	8.77	69
United Arab Emirates	0.00	0	22.59	5.41	17.18	64
UK	0.00	0	8.92	5.55	3.37	82
USA	10.26	0	11.99	2.20	9.79	1
Uruguay	6.75	0	11.79	5.39	6.41	26
Venezuela	14.12	0	16.03	3.44	12.59	5
Virgin Islands USA	11.00	11	29.35	15.35	14.00	105
Virgin Islands UK	0.00	0	26.37	5.40	20.97	29

Column (1) is the average withholding tax on dividends in the host countries
Column (2) presents the minimum withholding tax rates on dividends
Column (5) is the difference between columns (3) and (4). The results in both columns follow from the network analysis
The high rankings of the USA, Japan, and India are explained by their relatively high inbound repatriation tax rates. Given the size of their economies, these rates contribute to higher outbound rates for the other countries and thus the three stand out with lower average outbound rates

Appendix B3: Centrality measures

	BTWNS		OCCUR		STRCT		UNWTD	
	%	Rank	%	Rank	%	Rank	%	Rank
Albania	0.2	55	14.9	57	0.1	47	0.2	57
Algeria	0.0	81	0.0	81	0.0	84	0.0	83
Angola	0.0	74	0.0	82	0.0	85	0.0	84
Argentina	0.0	75	0.0	83	0.0	86	0.0	85
Aruba	0.0	73	0.0	84	0.0	87	0.0	74
Australia	1.3	35	27.1	50	0.1	44	0.2	56
Austria	2.1	22	46.8	26	0.6	25	3.0	18
Azerbaijan	0.1	59	5.8	66	0.0	64	0.0	69
Bahamas	0.8	41	35.7	46	0.0	53	2.1	30
Bahrain	0.0	76	0.0	85	0.0	88	0.0	76
Barbados	0.2	56	16.5	54	0.1	43	1.9	37
Belarus	0.1	60	13.1	60	0.0	60	0.1	58
Belgium	1.1	38	36.5	38	0.1	45	1.5	42
Bermuda	0.8	42	35.8	40	0.0	54	2.1	31
Botswana	0.0	77	0.0	86	0.0	89	0.0	86
Brazil	0.1	61	3.4	77	0.0	74	0.1	65
Brunei Darussalam	2.6	18	65.1	9	1.2	14	2.8	20
Bulgaria	2.0	23	47.8	25	0.5	30	2.2	29
Canada	0.1	62	5.8	67	0.1	48	0.2	55
Cayman Islands	0.8	43	35.8	41	0.0	55	2.1	32
Chile	0.0	78	7.0	65	0.0	71	0.0	78
China	0.0	79	5.7	70	0.0	70	0.0	75
Colombia	1.9	27	60.5	15	0.3	33	2.3	23
Costa Rica	0.0	80	0.0	87	0.0	90	0.0	87
Croatia	0.4	52	29.5	48	0.2	39	0.6	51
Curacao	1.7	30	61.8	14	0.3	34	2.7	21
Cyprus	4.5	10	59.2	17	1.8	10	8.1	3
Czech Republic	1.4	33	41.3	32	0.2	35	1.4	45
Denmark	4.0	13	55.4	19	1.1	15	2.2	25
Dominican Rep.	0.1	63	5.8	68	0.0	77	0.0	71
Ecuador	0.0	82	0.0	88	0.0	91	0.0	88
Egypt	0.4	51	39.8	33	0.0	61	0.4	53
Equatorial Guinea	0.0	84	0.0	89	0.0	92	0.0	89
Estonia	6.7	4	76.0	3	3.1	5	6.3	7
Finland	4.7	9	63.4	11	1.6	12	2.2	28
France	1.5	32	36.4	39	0.5	31	2.2	27
Gabon	0.0	83	0.0	90	0.0	93	0.0	90

	BTWNS		OCCUR		STRCT		UNWTD	
	%	Rank	%	Rank	%	Rank	%	Rank
Germany	0.8	44	29.3	49	0.1	51	1.0	49
Greece	1.7	31	45.2	28	0.4	32	1.4	46
Guernsey	0.8	45	35.8	42	0.0	56	2.1	33
Hong Kong	2.3	19	42.8	29	0.9	20	4.1	12
Hungary	6.2	5	76.9	1	2.3	7	6.9	5
Iceland	1.9	28	48.3	24	0.6	26	1.4	44
India	0.3	53	2.6	78	0.1	46	0.3	54
Indonesia	0.1	64	8.9	62	0.0	69	0.0	70
Ireland	5.6	7	70.1	5	2.9	6	3.8	14
Isle of Man	0.8	46	35.8	43	0.0	57	2.1	34
Israel	0.1	65	9.8	61	0.0	68	0.1	62
Italy	0.8	47	32.2	47	0.1	52	0.9	50
Jamaica	0.0	85	4.3	74	0.0	72	0.1	63
Japan	0.0	87	0.0	91	0.0	94	0.0	91
Jersey	0.8	48	35.8	44	0.0	58	2.1	35
Jordan	0.0	86	0.0	92	0.0	95	0.0	92
Kazakhstan	0.1	66	5.8	69	0.0	62	0.0	79
Korea Republic	0.1	67	7.4	63	0.0	66	0.0	66
Kuwait	0.0	88	0.0	93	0.0	96	0.0	93
Latvia	2.0	24	49.3	22	0.6	23	1.7	40
Lebanon	0.0	89	0.1	79	0.0	97	0.0	80
Libya	0.0	90	0.0	94	0.0	98	0.0	94
Liechtenstein	1.0	39	37.3	36	0.2	40	2.3	22
Lithuania	2.3	20	51.4	20	0.8	21	1.7	41
Luxembourg	8.4	2	76.4	2	7.6	2	5.2	10
Macao	0.0	91	0.0	95	0.0	99	0.0	95
Malaysia	4.0	14	62.7	13	1.0	17	5.6	8
Malta	4.3	12	60.5	16	1.7	11	6.6	6
Mauritius	1.3	36	36.8	37	0.2	37	3.4	15
Mexico	0.0	92	0.0	96	0.0	100	0.0	96
Mongolia	0.2	57	13.6	59	0.0	65	0.1	60
Namibia	0.0	93	0.0	97	0.0	78	0.0	97
Netherlands	7.7	3	66.7	8	3.5	4	9.6	2
New Zealand	0.2	58	14.0	58	0.0	67	0.1	64
Nigeria	0.0	94	0.0	98	0.0	79	0.0	98
Norway	2.0	25	42.7	30	0.6	24	1.9	38
Oman	2.3	21	63.4	12	0.5	29	3.1	16
Pakistan	0.0	95	0.0	99	0.0	80	0.0	99
Panama	0.0	96	0.0	100	0.0	101	0.0	100

	BTWNS		OCCUR		STRCT		UNWTD	
	%	Rank	%	Rank	%	Rank	%	Rank
Peru	0.0	97	0.0	101	0.0	75	0.0	101
Philippines	0.0	98	0.0	102	0.0	81	0.0	102
Poland	1.4	34	39.8	34	0.2	38	1.9	39
Portugal	1.0	40	37.4	35	0.1	42	1.0	48
Puerto Rico	0.0	99	0.0	103	0.0	82	0.0	103
Qatar	1.3	37	22.6	51	1.1	16	3.1	17
Romania	1.8	29	41.6	31	0.5	28	2.2	26
Russian Federation	0.1	68	5.1	72	0.1	50	0.0	67
Saudi Arabia	0.1	69	15.3	56	0.6	27	0.1	59
Serbia and Mont.	0.1	71	5.2	71	0.2	41	0.0	73
Seychelles	0.0	101	0.0	104	0.0	102	0.0	104
Singapore	6.1	6	67.4	7	4.3	3	7.4	4
Slovak Republic	5.3	8	68.2	6	2.1	8	4.7	11
Slovenia	2.0	26	49.3	23	0.7	22	1.5	43
South Africa	0.1	72	16.1	55	0.0	76	0.1	61
Spain	3.6	15	45.7	27	2.0	9	5.4	9
Suriname	0.0	100	0.0	105	0.0	103	0.0	105
Sweden	4.5	11	57.6	18	1.4	13	2.3	24
Switzerland	3.2	16	50.9	21	1.0	19	3.0	19
Taiwan Province	0.0	104	4.9	73	0.0	83	0.0	77
Thailand	0.0	102	3.5	76	0.0	104	0.0	82
Trinidad and Tob.	0.1	70	7.3	64	0.0	63	0.4	52
Tunisia	0.0	103	0.0	106	0.0	105	0.0	106
Turkey	0.6	50	19.6	53	0.1	49	0.0	68
Ukraine	0.3	54	20.9	52	0.2	36	1.1	47
United Arab Emirates	2.8	17	64.2	10	1.0	18	4.1	13
UK	13.4	1	74.4	4	13.8	1	12.8	1
USA	0.0	106	0.1	80	0.0	106	0.0	81
Uruguay	0.0	105	0.0	107	0.0	107	0.0	107
Venezuela	0.0	107	3.7	75	0.0	73	0.0	72
Virgin Islands USA	0.0	108	0.0	108	0.0	108	0.0	108
Virgin Islands UK	0.8	49	35.8	45	0.0	59	2.1	36

Appendix C1: The adapted shortest path algorithm

This appendix describes the adaptations to the standard shortest path algorithm required to fit the optimal tax routes. Although this material is mainly theoretical, we also include the results of computations to motivate the choice we made concerning the credit method in conduit situations.

The standard all-pair shortest path problem (APSP) is solved with the Floyd–Warshall algorithm.[38] The core of this algorithm is the next comparison, where d_{ij}^m is the length of the shortest path from i to j allowing only the first m vertices (countries) as intermediate stations.

$$d_{ij}^m = \min\{d_{im}^{m-1} + d_{mj}^{m-1}, d_{ij}^{m-1}\}$$

The algorithm is initialized with the distance matrix, which contains all the relevant information ($D^0 = D$). By consecutively allowing an additional vertex as intermediate station, the length of the shortest path over the whole network is computed for all possible pairs ($S = D^{\mathrm{N}}$). The efficiency of the algorithm is that with a fixed and limited number of additions and comparisons, each of the order N^3, hence not polynomial, it completes the job.

The core comparison of the algorithm reflects that in the world of transportation distances simply can be added. This is obviously not the case for tax rates, as the base for taxation with a second rate, are the profits after the first tax. The first adaptation makes the switch to the multiplicative character of taxes and corresponds with *deduction* as the method for double taxation relief.

$$d_{ij}^m = \min\{d_{im}^{m-1} + d_{mj}^{m-1} - d_{im}^{m-1} d_{mj}^{m-1}, d_{ij}^{m-1}\}$$
$$\text{or} \quad d_{ij}^m = \min\{1 - (1 - d_{im}^{m-1})(1 - d_{mj}^{m-1}), d_{ij}^{m-1}\}$$

The tax rates include the non-resident withholding taxes, which are given for a country pair, i.e. from i to j. The country-specific corporate income taxes (CITs) are included as part of the compounded tax distances for inward income flows (see Sect. 2). This is the second adaption.

There is a convenient consequence of including the CIT of a home country in the tax distances applying to its inward flows. For countries with *exemption* as their double tax relief method, it amounts to having a CIT of zero. Their actual CIT only matters when these countries are the initial host on a repatriation path.

In general, for any tax route, with an initial host $k = 1$ and final destination $k = \mathrm{n}$, the full combined effective tax rate equals $1 - (1 - t_1)\left(\prod_{k=2}^{n}(1 - d_{k-1,k})\right)$. Here t_1 denotes the CIT of country 1, and $d_{1,2} = w_{1,2}$ is the bilateral withholding tax rate from 1 to 2. The other tax distances are either the bilateral withholding tax rates, $d_{k-1,k} = w_{k-1,k}$, when country k applies *exemption*, or they include the CIT of the intermediate home country k, $d_{k-1,k} = 1 - (1 - w_{k-1,k})(1 - t_k)$, when it applies *deduction* as double tax relief. The adapted Floyd–Warshall takes care of the product

[38] See, for instance, Minieka (1978).

of the tax distances, in which the order is inconsequential.[39] Thus, a basic method is defined, with a deduction "metric", covering both *deduction* and *exemption* as double tax relief methods.[40]

Incorporating the *credit* method introduces a complexity which requires two final adaptations. The question is which taxes can be credited against the corporate tax in the final or intermediate, home country. Roughly three possibilities can be identified: (1) all taxes paid on the preceding tax route are credited, (2) only the taxes actually paid in the last preceding jurisdiction are credited, and (3) the nominal CIT rate of the last preceding jurisdiction is credited, whether this CIT is paid or not. In the latter case also the withholding tax would be credited.[41]

The option of crediting the nominal CIT of the last preceding country, option iii), has the advantage that nothing needs to be known of the route before that last country visited. Moreover, it fits into the method described above, with the definition of tax distance also given in the main text:

$$d_{k-1,k}(\text{credit}) = \max\{w_{k-1,k},\ (t_k - t_{k-1})/(1 - t_{k-1})\}$$

The first option may be most in line with the philosophy of the credit method, i.e. capital export neutrality. In practice it may be difficult, or undesirable, to account for all the accumulated taxes paid on a tax route. These total taxes include the treatment of the CIT of the evaluated jurisdiction. However, it must be realized that the treatment of the CIT in the jurisdiction under consideration is based on the initial distance matrix, so that the credit is based on the *nominal* tax rate of the previous jurisdiction on the path instead of the *actual* total taxes paid. This excludes implementing the first two options.

Acknowledging that the practice of the credit method is complex and that we have no structural information on the actual operation of the method we decided on the next implementation: we let the world average CIT be credited,[42] in combination with the actual withholding tax of the last conduit jurisdiction preceding the parent jurisdiction with the credit method.

We believe this solution is the most realistic among possible alternatives. A first alternative approach would be to assume that no taxes at all were paid so that no credits are applied. This would seriously *under*estimate the potential reduction of the tax burden for MNEs by treaty shopping. We find a world average optimal dividend repatriation tax of 9.87% instead of 5.75%. On the other hand, taking the statutory CIT of a conduit country as the basis for tax credit would *over*estimate the potential reduction as this CIT is not likely to be paid in full because of double tax relief. Moreover, with this alternative, the optimal tax routes would make use of those con-

[39] This is the commutative property.

[40] Strictly speaking it is not a metric, since, for instance, the property of symmetry is not satisfied.

[41] This is the indirect tax credit system; see below.

[42] Specifically, the world average CIT excludes the one of the countries under consideration.

duit countries that apply exemption but which have a high nominal CIT, which then would be credited. The world average optimal rate found for this alternative is 3.26%, reflecting the underestimation compared to the 5.75% when the world average CIT is credited.

Furthermore, it must be observed that the world average tax rate is only applied in those conduit situations where a jurisdiction with the credit method follows a conduit country. When the last preceding jurisdiction is the starting point of a tax route, the corporate income tax is paid in the initial host and is credited in the next stop of a tax route.

This gives rise to the final adaptation of the shortest path algorithm. Let d_{ij} denote the usual tax distance between i and j when i is the first node of a path, and let p_{ij} denote the distance between i and j when i is an intermediate node on a path. This second distance incorporates the assumption dealing with the credit method.

$$p_{k-1,k}(\text{credit}) = \max\{\ w_{k-1,k},\ (t_k - \bar{t})/(1-\bar{t})\}, \quad \text{with } \bar{t} : \text{world average CIT}$$

Let p_{ij}^N be the output of the Floyd–Warshall algorithm with the deduction "metric" applied to distances for intermediate stations. Thus, all shortest distances are known for the inner work of tax routes, i.e. when the first vertex of the route eventually is the second. Then the outer work of initial vertices (jurisdictions) can be added as follows.

$$d_{ij}^N = \min\{d_{ij}, \min_m(d_{im} + p_{mj}^N)\}$$

Finally, we briefly discuss two relief options that we have not covered. Instead of allowing both the corporate tax of the host country and the withholding tax to be credited, some countries only allow the withholding tax to be credited against their corporate tax. The latter method is referred to as a direct foreign tax credit whereas the former is the indirect tax credit method. For conduit situations we use the indirect credit method. The direct credit method could also easily have been implemented; it suffices to define the tax distance for i as a first vertex of a tax route; see below. We have, however, not collected information on countries applying direct rather than indirect credits.

$$d_{ij}(\text{direct credit}) = \max\{w_{ij}, t_j\}$$

Some countries provide no-relief-at-all for double taxation; the combined tax rate for a direct route equals:

$$t_{SP}^e(\text{no relief}) = t_S + w_{SP} - t_S w_{SP} + t_P$$

In conduit situations problems similar to those with indirect credits occur, although no-relief-at-all is not likely to occur in conduit situations. Nevertheless, we have not covered it.

Generating all shortest paths, and all those within range

The Floyd–Warshall algorithm is an efficient method to compute the value of the strict shortest paths for all pairs of nodes of a network. With a small addition to the algorithm, the so-called penultimate vertex matrix (PVM) can be maintained. Upon completion of the Floyd–Warshall algorithm, shortest paths for all country pairs can be reconstructed from this matrix. The PVM method generates only a single strict shortest path for a given pair. We, however, require all shortest paths of a given pair, to be able to compute centrality measures. In addition, we are also interested in those paths for a given pair with a length that is within a prespecified admissible range on top of the value of the strictly shortest path. The PVM method is not suitable for generating all those relevant paths.

Instead we implement a branch and bound method. The branching consists of a full, depth-first enumeration of all possible combinations. The bounding is accomplished with the values of the strict shortest paths which were computed with the Floyd–Warshall algorithm, executed beforehand. This implementation is a brute-force approach. It is only possible because the relevant paths are not too long, with a sequence of five or six countries as a maximum. Polak (2015) describes a relative efficient implementation of the brute-force method.

Appendix C2: The betweenness centrality measure and flows

Country—weights w_i are defined as $w_i = GDP_i / \sum_k GDP_k$. Of course, $\sum_i w_i = 1$.
Double GDP—weights on the flows (i, j) are:

$$w_{ij} = w_i \frac{GDP_j}{\sum_k GDP_k - GDP_i} = w_i w_j \frac{1}{(1 - w_i)}.$$

By construction $\sum_i \sum_{j \neq i} w_{ij} = 1$. The weights are the flows when 1 euro or dollar is run through the network; they are the shares of the total of the flows.

The measure of *betweenness centrality* for vertex k, B_k, is computed from the number of times vertex k is on a relevant path from i to j, excluding k as start and end point, n_{kij}, as a share in the total number of relevant paths from i to j, N_{ij}, and then these fractions are weighted over all pairs i and j. $B_k = \sum_{i \neq k} \sum_{j \neq i,k} w_{ij} \frac{n_{kij}}{N_{ij}}$

The assumption is that each of the relevant paths between i and j takes the same share, i.e. $1/N_{ij}$, of the total flow of the pair ij, whose share is w_{ij}. The weights can be given the interpretation of the relative size of the undiverted bilateral flows. By construction *betweenness centrality* then measures the share of total undiverted flows that run through a vertex, excluding all the flows that start or end at the given vertex k.

Table 15 Summary statistics—gravity variables

Variable	Mean	Std. dev.	Min	Max
Log GDP	12.07	1.98	6.80	16.63
Average institutional quality	0.57	0.83	− 1.37	1.86
Log distance	8.55	0.94	4.09	9.89
Contiguity dummy	0.03	0.17	0.00	1.00
Common language dummy	0.11	0.31	0.00	1.00
Colony dummy	0.02	0.15	0.00	1.00
Dummy former common country	0.01	0.09	0.00	1.00
Dummy both OECD countries	0.15	0.36	0.00	1.00
Common regional trade agreement	0.26	0.44	0.00	1.00
Common legal system	0.27	0.44	0.00	1.00

Number of observations: 6422

Appendix D1: Gravity variables

This appendix presents summary statistics and results for the gravity variables. GDP and distance (in kilometres) are measured in logs. The indicator for institutional quality is an average of the Worldwide Governance Indicators (World Bank 2016), each ranked between − 2.5 and 2.5. A higher value implies higher institutional quality. The geographic variables are downloaded from CEPII. Countries are only in 3% of the cases close neighbours. Of the country pairs 11% has a common language, 2% has a colonial relation, 26% a common regional trade agreement, and 27% has a common legal system (Table 15).

The coefficients for the gravity variables in the OLS regressions are found in Table 16. GDP in the host and home country has a positive impact on the bilateral FDI stock. This is also the case for institutional quality in both countries. Benassy-Quere et al. (2007) and Egger et al. (2009), among others, use GDP per capita instead of institutional quality. Both variables are heavily correlated. We have chosen for institutional quality because the explanatory power is higher in our sample. Distance has the expected negative effect on the size of the bilateral FDI stock; a common language increases bilateral investments, as is also the case for contiguity, a colonial relationship in the past, OECD membership of both countries and a common legal system. A regional trade agreement RTA and a former common country also increase FDI, but the coefficients of these variables are not significant.

The results for the gravity variables in the robustness analysis are given in Table 17.

Table 16 Explaining FDI stocks with indicators for FDI diversion—gravity variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log(GDP_home)	0.761***	0.759***	0.759***	0.769***	0.759***	0.759***	0.708***
	(0.0189)	(0.0193)	(0.0193)	(0.0191)	(0.0192)	(0.0192)	(0.0193)
Log(GDP_host)	0.652***	0.637***	0.636***	0.641***	0.632***	0.631***	0.627***
	(0.0188)	(0.0188)	(0.0188)	(0.0187)	(0.0187)	(0.0187)	(0.0186)
INST_home	1.163***	1.072***	1.083***	1.115***	1.044***	1.054***	1.037***
	(0.0395)	(0.0399)	(0.0403)	(0.0386)	(0.0442)	(0.0445)	(0.0446)
INST_host	0.323***	0.338***	0.353***	0.363***	0.268***	0.284***	0.306***
	(0.0420)	(0.0411)	(0.0416)	(0.0417)	(0.0449)	(0.0454)	(0.0460)
Log distance	−0.879***	−0.730***	−0.728***	−0.755***	−0.710***	−0.708***	−0.746***
	(0.0398)	(0.0413)	(0.0413)	(0.0408)	(0.0416)	(0.0416)	(0.0418)
Dummy contiguity	0.600***	0.730***	0.735***	0.709***	0.803***	0.807***	0.801***
	(0.188)	(0.187)	(0.186)	(0.185)	(0.187)	(0.186)	(0.184)
Dummy language	1.376***	1.495***	1.493***	1.490***	1.528***	1.526***	1.436***
	(0.120)	(0.119)	(0.119)	(0.119)	(0.119)	(0.119)	(0.120)
Dummy colony	1.543***	1.460***	1.453***	1.467***	1.324***	1.318***	1.279***
	(0.203)	(0.194)	(0.194)	(0.195)	(0.200)	(0.200)	(0.204)
Dummy former common country	0.384	0.465	0.466	0.434	0.457	0.459	0.462
	(0.305)	(0.317)	(0.316)	(0.312)	(0.319)	(0.318)	(0.314)
Dummy OECD pair	0.630***	0.574***	0.556***	0.568***	0.583***	0.566***	0.513***
	(0.111)	(0.108)	(0.108)	(0.107)	(0.107)	(0.107)	(0.108)
Dummy RTA	0.0491	0.0714	0.0729	0.0976	0.0390	0.0407	−0.0674
	(0.0843)	(0.0821)	(0.0821)	(0.0819)	(0.0827)	(0.0827)	(0.0835)
Dummy common legal system	0.223***	0.263***	0.262***	0.268***	0.267***	0.266***	0.228***
	(0.073)	(0.072)	(0.072)	(0.072)	(0.072)	(0.072)	(0.072)

The regressions are estimated with OLS. There are 6422 observations. Robust standard errors are in parentheses, ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$

Table 17 Robustness analyses—gravity variables

Robustness	(1) Fixed effect	(2) PPML	(3) OECD sample	(4) Tax haven	(5) Occurence	(6) Strict	(7) Unweighted
Log(GDP_home)		0.478***	1.190***	0.779***	0.768***	0.753***	0.758***
		(0.0778)	(0.0325)	(0.0194)	(0.0195)	(0.0192)	(0.0192)
Log(GDP_host)		0.527***	0.815***	0.634***	0.641***	0.629***	0.626***
		(0.0694)	(0.0289)	(0.0193)	(0.0188)	(0.0188)	(0.0186)
INST_home		0.880***	1.365***	1.069***	1.042***	1.049***	0.991***
		(0.105)	(0.0887)	(0.0446)	(0.0469)	(0.0413)	(0.0428)
INST_host		0.445***	0.664***	0.296***	0.233***	0.318***	0.230***
		(0.112)	(0.0765)	(0.0457)	(0.0493)	(0.0424)	(0.0440)
Log distance	−0.947***	−0.316***	−0.822***	−0.705***	−0.697***	−0.710***	−0.694***
	(0.0436)	(0.100)	(0.0610)	(0.0415)	(0.0421)	(0.0413)	(0.0411)
Dummy contiguity	0.300*	0.265	0.0637	0.836***	0.795***	0.823***	0.906***
	(0.175)	(0.232)	(0.272)	(0.182)	(0.187)	(0.186)	(0.184)
Dummy language	0.801***	0.551***	0.588***	1.652***	1.562***	1.502***	1.549***
	(0.112)	(0.209)	(0.185)	(0.117)	(0.120)	(0.119)	(0.119)
Dummy colony	1.249***	0.0840	1.245***	1.081***	1.385***	1.249***	1.117***
	(0.182)	(0.219)	(0.240)	(0.200)	(0.194)	(0.205)	(0.198)
Dummy former common country	0.336	1.611***	0.834**	0.388	0.445	0.473	0.435
	(0.304)	(0.325)	(0.388)	(0.324)	(0.320)	(0.317)	(0.322)
Dummy OECD pair	0.713***	0.294	−0.388***	0.567***	0.581***	0.560***	0.604***
	(0.105)	(0.180)	(0.146)	(0.110)	(0.108)	(0.107)	(0.107)
Dummy RTA	0.510***	−0.882***	0.307***	0.0702	0.0231	0.0508	−0.00920
	(0.0775)	(0.213)	(0.118)	(0.0827)	(0.0826)	(0.0823)	(0.0825)
Dummy common legal system	0.365***	−0.0589	0.664***	0.261***	0.264***	0.268***	0.253***
	(0.0600)	(0.199)	(0.105)	(0.0717)	(0.0717)	(0.0718)	(0.0711)

The regressions are estimated with OLS and robust Huber–White standard errors, except for column (2). There are 6422 observations, except in the OECD sample where we have 2783 observations. ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$

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